Algebra 1 Notes

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Fall 2025

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Algebra 1

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2025 Fall
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$$\mathbb{N},\ \mathbb{Z},\ \mathbb{Q},\ \mathbb{R},\ \mathbb{C}.$$

Question. How can we solve

$$ax^3 + bx^2 + cx + d = 0\,?$$

Without loss of generality, solve

$$x^3 + bx^2 + cx + d = 0.$$

Substituting $x - b/3$ into x, the equation becomes

$$x^3 + px + q = 0.$$

Solution Let $x = u + v$. Expand:

$$(u^3 + v^3 + q) + (3uv + p)(u + v) = 0.$$

If both are 0, then it is a solution for us:

$$\begin{cases} u^3 + v^3 = -q, \\ uv = -\dfrac{p}{3} \end{cases} \qquad \Longleftrightarrow \qquad u^3 v^3 = -\frac{p^3}{27}.$$

Hence u^3, v^3 are solutions for the quadratic

$$y^2 + qy - \frac{p^3}{27} = 0,$$

and

$$u^3, v^3 = \frac{-q \pm \sqrt{q^2 + 4p^3/27}}{2}.$$

Thus one solution for x is

$$x = \sqrt[3]{\frac{-q + \sqrt{q^2 + 4p^3/27}}{2}} + \sqrt[3]{\frac{-q - \sqrt{q^2 + 4p^3/27}}{2}}.$$

Side note. There is no bijection $f : A \to \mathcal{P}(A)$. Suppose

$$S := \{a \in A \mid a \notin f(a)\}.$$

There exists $a \in A$ such that $f(a) = S$. Then

$$a \in S \Rightarrow a \notin S, \qquad a \notin S \Rightarrow a \in S.$$

Question. What if

$$q^2 + 4\frac{p^3}{27} < 0\,?$$

Answer. Construction of $\mathbb{C}$:

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}, \qquad i^2 = -1,$$

where a is the real part and bi is the imaginary part. Or, as in Rudin,

$$\mathbb{C} := \{(a, b) \mid a, b \in \mathbb{R}\},$$

with

$$(a_1, b_1)(a_2, b_2) := (a_1a_2 - b_1b_2,\ a_1b_2 + a_2b_1),$$

together with multiplication and plus addition.

$\mathbb{N}$ axioms

Definition

$$\mathbb{N} = \{\text{distinguished element } 0 \text{ and successor function } S : \mathbb{N} \to \mathbb{N}\},$$

with S satisfying the theorem of induction:

$$\begin{cases} P(0), \\ P(n) \Rightarrow P(S(n)) \end{cases} \qquad \Longrightarrow \quad P(n) \text{ true}, \quad \forall n \in \mathbb{N}.$$

Note. If $A \subset \mathbb{N}$, $0 \in A$, and $S(A) \subset A$, then $A = \mathbb{N}$.

Definition (Addition)

$$+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}, \qquad (a, b) \longmapsto a + b,$$

with, for all $a, b \in \mathbb{N}$,

$$\begin{cases} a + 0 := a, \\ a + S(b) := S(a + b). \end{cases}$$

(recursive definition)

Side note: Well-Ordering Principle of $\mathbb{N}$ by induction.

$$\begin{gathered} n = 1 \in S \Rightarrow S \text{ has least element}, \\ n \in S \Rightarrow S \text{ has least element}, \\ \Downarrow \\ n + 1 \in S \Rightarrow S \text{ has least element}; \\ \{1, \ldots, n\} \cap S = \varnothing \Rightarrow n + 1 \text{ is least element}. \end{gathered}$$

Thus the well-ordering principle of $\mathbb{N}$ follows by induction.

Fact.

$0 + a = a$.

Proof

Base case:

$$0 + 0 = 0.$$

Induction step:

$$\begin{aligned} 0 + S(a) &= S(0 + a) \\ &= S(a) \qquad \text{induction hypothesis} \end{aligned}$$

Fact

$$a + S(b) = S(a) + b, \qquad \forall a, b \in \mathbb{N}.$$

Proof

By induction on any variable.

Fact

$$a + b = b + a.$$

Proof

Induction on b.
Base case: $b = 0$, and

$$a + 0 = a = 0 + a$$

by Fact 1 and definition.
Induction step $b \Rightarrow S(b)$:

$$\begin{aligned} a + S(b) &= S(a + b) = S(b + a) \\ &= b + S(a) = S(b) + a \end{aligned}$$

by the previous fact.

Definition (multiplication)

$$\begin{cases} a \times 0 = 0, \\ a \times S(b) = a \times b + a, \end{cases} \qquad \times : \mathbb{N} \times \mathbb{N} \to \mathbb{N}.$$

Fact

$\times$ is commutative and multiplicative.

Definition

$(\mathbb{Z}, \times, +)$.
For $+$: commutative, associative, neutral element 0,

$$0 + a = a + 0 = a,$$

and an inverse element: for every $a \in \mathbb{Z}$, there exists $a^* \in \mathbb{Z}$ such that

$$a + a^* = 0.$$

Thus $(\mathbb{Z}, +, 0)$ is a commutative group/Abelian group.

Group = associative, inverse, identity element.

For $\times$: same, same, neutral element 1,

$$1 \times a = a \times 1 = a, \qquad \forall a.$$

This is a commutative monoid.

Distributative law.

$$(a + b) \times c = a \times c + b \times c.$$

Therefore

$$(\mathbb{Z}, +, \times, 0, 1)$$

(and the distributative law) is a commutative ring.

Well-Ordering Principle of $\mathbb{N}$

Definition

Inductively define order on $\mathbb{N}$ by

$$\begin{cases} a \leq a, & \forall a \in \mathbb{N}, \\ a \leq b \Rightarrow a \leq S(b). \end{cases}$$

Extend to $\mathbb{Z}$ by

$$a \leq b \Rightarrow -b \leq -a, \qquad \forall a, b \in \mathbb{N}.$$

Well-Ordering Principle of $\mathbb{N}$. For every $S \subset \mathbb{N}$ such that $S \neq \varnothing$, S has a least element.

Proof

Prove the $P(n)$:

$$n \in \mathbb{N} \Rightarrow S \text{ has a least element}$$

by mathematical induction. Then the well-ordering principle of $\mathbb{N}$ follows.

Proof of strong/transfinite induction. Suppose $P(n)$ such that

$$P(\text{base case}) \text{ is true},$$
$$P(k \le n) \text{ true} \Rightarrow P(n+1) \text{ true}.$$

Let

$$Q(n) := \big(P(k) \text{ is true}, \ \forall k \le n\big).$$

Then the two conditions imply $Q(n)$ is true by mathematical induction; hence $Q(n)$ is true for every n, and therefore $P(n)$ is true for every n.

Or use strong induction to show $n \notin S$, $\forall n \in \mathbb{N}$, if S does not have a least element. Then $S = \varnothing$, a contradiction.

Division with remainder

Theorem

If $a, b \in \mathbb{Z}$ and $b \neq 0$, then there exist unique $q, r \in \mathbb{Z}$,

$$0 \le r < |b|, \qquad a = bq + r.$$

Proof

(Using induction on $|a|$ is doable.) Let

$$R = \{a - bq \ge 0 \mid q \in \mathbb{Z}\}.$$

Then $R \subset \mathbb{N}$ and $R \neq \varnothing$. Let r be the least element of R. Then $r < |b|$. If not, $r - |b|$ is the least element of R, a contradiction. Thus $r < |b|$; existence is proven.
For uniqueness, if

$$a = bq + r = bq' + r', \qquad r, r' < |b|,$$

then

$$b \mid r - r', \qquad r - r' = b(q' - q),$$

so $r = r'$, and then $q = q'$.

Divisibility

Definition

For $a, d \in \mathbb{Z}$, we say d divides a if there exists $k \in \mathbb{Z}$ such that

$$a = dk.$$

We write $d \mid a$.

Proposition

If $d \mid a$ and $d \mid b$, then

$$d \mid ma + nb.$$

Proposition

If $a \mid b$ and $b \mid a$, then $a = \pm b$.

Greatest Common Divisors

Definition

The gcd of $a, b \in \mathbb{Z}$ is the greatest integer that divides both a and b.

Theorem

Every common divisor of $a, b \in \mathbb{Z}$ divides $\gcd(a, b)$. There exist $m, n \in \mathbb{Z}$ such that

$$ma + nb = d.$$

Proof

Take the least positive element in

$$D := a\mathbb{Z} + b\mathbb{Z},$$

say d. Perform division:

$$a = qd + r, \qquad 0 \leq r < d.$$

Then $r \in D$ or $r = 0$, since

$$r = a - qd = a - q(ma + nb) \in D.$$

Hence $r = 0$, otherwise d is not the least element. Therefore $d \mid a$. Similarly, $d \mid b$.
If $d' \mid a, b$, then

$$d' \mid ma + nb = d,$$

so $d' \leq d$.

Corollary

If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.

Proof

$$\begin{aligned} ma + nb &= 1, \\ mac + nbc &= c, \\ mac + nqa &= c, \\ a(mc + nq) &= c, \end{aligned}$$

hence $a \mid c$.

Euclidean Algorithm: Computing Greatest Common Divisors

Corollary

To find the gcd of a, b, pick the larger one and divide:

$$a = bq + r, \qquad \gcd(a, b) = \gcd(b, r).$$

Then continue

$$a \div b = q_1 \cdots r_1.$$

Corollary (continued)

The Euclidean algorithm continues as

$$\begin{aligned} b \div r_1 &= q_2 \cdots r_2, \\ r_1 \div r_2 &= \cdots r_3, \\ r_2 \div r_3 &= \cdots r_4, \\ &\vdots \\ \gcd \quad &\cdots 0. \end{aligned}$$

Example

$$\begin{aligned} 18 \div 10 &= 1 \cdots 8, \\ 10 \div 8 &= 1 \cdots 2, \\ 8 \div 2 &= 4 \cdots 0, \end{aligned}$$

so $\gcd(18, 10) = 2$.

Theorem

There exist infinitely many primes.

Proof

By contradiction: $p_1 p_2 \cdots p_n + 1$, prime. (How? By the Fundamental Theorem of Arithmetic below.)

Fundamental Theorem of Arithmetic

1. For every $n \in \mathbb{Z}$, n can be written as a product of primes:

$$n = \pm p_1^{e_1} \cdots p_m^{e_m},$$

   where the p_i's are distinct primes.
2. The expression is unique upon rearrangement.

Proof of (1)

Let

$$S := \{n \mid n \text{ cannot be written as } \cdots\} \cap \mathbb{N}.$$

Suppose $S \neq \varnothing$. Then there is a least element $n_0 \in S$. If n_0 is prime, then $n_0 \notin S$. Suppose

$$n_0 = ab, \qquad a, b < n_0.$$

Then $a, b \notin S$, so a, b are products of primes, whence $n_0 \notin S$, a contradiction. Therefore $S = \varnothing$.

Gauss' Lemma.

Let p be a prime. If $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof

Suppose $p \nmid a$. Then

$$(p, a) = 1 \Rightarrow p \mid b$$

by the previous corollary.

[extended to case n with mathematical induction]

Proof of (2)

Suppose

$$n = p_1^{e_1} \cdots p_n^{e_n} = q_1^{s_1} \cdots q_r^{s_r}.$$

Then p_1 divides something from the right, so $p_1 \mid q_1$, and therefore $p_1 = q_1$. Proceed by strong induction (holds for $\leq n$; prove $n + 1$).

Congruences

Fix an integer $m > 1$.

Definition (Gauss)

Two integers are said to be *congruent modulo* m if m divides $b - a$. We write

$$a \equiv b \pmod{m} \qquad \text{or} \qquad a \equiv_m b.$$

Property. (Proof omitted.)

1. Reflexivity: $a \equiv_m a$.
2. Symmetry: $a \equiv_m b \Rightarrow b \equiv_m a$.
3. Transitivity:

$$a \equiv_m b, \quad b \equiv_m c \quad \Rightarrow \quad a \equiv_m c.$$

Conclusion. $\equiv_m$ is an equivalence relation on $\mathbb{Z}$.

Definition

The quotient set

$$\mathbb{Z}/\equiv_m$$

is the set of equivalence classes in $\mathbb{Z}$ under $\equiv_m$. We denote the equivalence class of $j \in \mathbb{Z}$ by $[j]_m$:

$$[j]_m := \{j + km \mid k \in \mathbb{Z}\}.$$

Given $a, b \in \mathbb{Z}$,

$$[a]_m = [b]_m \iff a \equiv_m b.$$

Lemma.

$\mathbb{Z}/\equiv_m$ is finite and

$$|\mathbb{Z}/\equiv_m| = m.$$

Proof

Given $N \in \mathbb{Z}$, we have

$$N = qm + r, \qquad 0 \leq r \leq m - 1.$$

Thus

$$N \equiv_m r \quad \Rightarrow \quad [N]_m = [r]_m.$$

Note. Therefore

$$\mathbb{Z}/\equiv_m = \{[0], [1], \ldots, [m-1]\}.$$

Quotients. In quotients, there can be multiple representatives. For example,

$$[0] = [m] = [-m] = [2m] = \cdots.$$

Consequently, $\equiv_m$ is compatible with $+$ and $\times$.

Proposition

If

$$a \equiv_m a', \qquad b \equiv_m b',$$

then

$$a + b \equiv_m a' + b', \qquad ab \equiv_m a'b'.$$

Notation.

$$\mathbb{Z}/\equiv_m := \mathbb{Z}/m\mathbb{Z}.$$

Proof

(last proposition) Trivial.

Corollary

$+$ and $\times$ on $\mathbb{Z}$ give well-defined $+$ and $\times$ on $\mathbb{Z}/m\mathbb{Z}$:

$$[a] + [b] := [a + b], \qquad [a][b] := [ab].$$

Corollary

$$(\mathbb{Z}/m\mathbb{Z}, +, \times)$$

is a ring.

Proof

Trivial.

Theorem

$\mathbb{Z}/p\mathbb{Z}$ is a field (p prime).

Proof

Let $[a] \neq 0$. Then

$$p \nmid a \quad \Rightarrow \quad (p, a) = 1.$$

Hence

$$ps + ak = 1,$$

and therefore

$$[ps + ak] = 1 = [ak].$$

Thus

$$[k] = [a]^{-1}.$$

Theorem

For every composite m, $\mathbb{Z}/m\mathbb{Z}$ is NOT a field.

Proof

Existence of 0 divisors implies inexistence of inverses.

Definition

A ring without zero divisors is an *integral domain*.

$$\text{Zero divisors: } ab = 0 \text{ where } a \neq 0,\ b \neq 0.$$

Cancellation Law

If R is an integral domain and $a \neq 0$, then

$$ax = ay \Rightarrow x = y.$$

Example Integral domains that are not fields (lack of multiplicative inverses):

$$\mathbb{Z}, \qquad \text{polynomial rings.}$$

Remark We have shown that

$$m \text{ is prime} \iff \mathbb{Z}/m\mathbb{Z} \text{ is a field} \iff \mathbb{Z}/m\mathbb{Z} \text{ is an integral domain (no zero divisors).}$$

Freshman's Dream

If p is prime, then for all $x, y \in \mathbb{Z}/p\mathbb{Z}$,

$$(x+y)^p = x^p + y^p.$$

Proof

binomial theorem expansion, and

$$[p] = [0].$$

Fermat's little theorem

Theorem

If p is a prime, then for every $a \in \mathbb{Z}$,

$$a^{p-1} \equiv_p 1 \qquad \text{or} \qquad a^p \equiv_p a.$$

Proof [1]

$\mathbb{Z}/p\mathbb{Z}$ is a group. Look at the order of $[a]$. Apply Lagrange's theorem to the order of the cyclic group

$$\langle a \rangle.$$

[2] Note $a \geq 0$ suffices:

$$\text{if } p = 2,\ [-1] = [1]; \qquad p \neq 2,\ \text{then do } -a.$$

Induction on a. For $a = 0$,

$$0^p \equiv 0.$$

For $a \Rightarrow a + 1$,

$$(a+1)^p \equiv a^p + 1^p \equiv a + 1,$$

by the lemma and the induction hypothesis.

Corollary

If there exists $a \in \mathbb{Z}$ such that

$$a^p \not\equiv a \pmod{p},$$

then p is not prime.

Converse. If

$$a^p \equiv a \pmod{p}, \qquad \forall a \in \mathbb{Z},$$

does it imply that p is prime? It does not imply it.

Such numbers are called Fermat pseudoprimes.

Example More efficient algorithms to check prime:

$$n = 15, \qquad a = 2.$$

Modulo 15,

$$2^{15} \equiv 4^7 \times 2 \equiv 16^3 \times 4 \times 2 \equiv 1^3 \times 4 \times 2 \equiv 8 \not\equiv 2.$$

[Calculate the congruence first, then use the contrapositive of Fermat's little theorem.]

Therefore 15 is not a prime by the contrapositive of Fermat's little theorem.

Example

$$M = 2^{2^{12}} + 1 \qquad \text{(Fermat's number)}.$$

By computer (and Fermat's little theorem),

$$3^M \not\equiv 3 \pmod{M}.$$

Wilson's Theorem

Theorem

$$p \text{ is prime} \iff (p-1)! \equiv -1 \pmod{p}.$$

Proof of ($\Rightarrow$)

Key observation: all the elements in

$$\bar{1}, \bar{2}, \ldots, \overline{p-1}$$

have a different inverse than itself, other than $\bar{1}$ and $\overline{p-1}$ ($= \bar{-1}$). Then

$$(p-1)! = \bar{1} \times \bar{2} \times \cdots \times \overline{p-1} = \bar{1} \times \overline{p-1} = \bar{-1}.$$

So

$$p - 1 \equiv -1.$$

($\Leftarrow$). If p is not a prime, then

$$(p-1)! \equiv 0 \in \mathbb{Z}/p\mathbb{Z},$$

a contradiction.

Corollary of Fermat's Little Theorem

If $a \in \mathbb{Z}/p\mathbb{Z}$ and $N \geq 1$, then a^N depends on N modulo $p-1$.

Proof

Write

$$N = (p-1)q + r, \qquad 0 \leq r < p-1.$$

Then

$$a^N = a^{(p-1)q+r} = 1^q a^r = a^r.$$

Example

$$2^{1234567} \equiv ? \pmod{11}.$$

Dividing the exponent,

$$1234567 \div \underbrace{(11-1)}_{p-1} = \text{something} \cdots 7.$$

Thus, in $\mathbb{Z}/11\mathbb{Z}$,

$$2^{1234567} \equiv 2^7 = 16 \times 4 \times 2 = 5 \times 4 \times 2 = 9 \times 2 = 7.$$

Rivest–Shamir–Adleman Public-Key Cryptosystem (Application)

Public key.

$$n = pq \qquad \text{(large primes)}.$$

Messages are elements of

$$\mathbb{Z}/n\mathbb{Z}.$$

Encryption Exponent e.

$$\gcd(e, (p-1)(q-1)) = 1.$$

The public information is n and e.

Encryption. For $M \in \mathbb{Z}/n\mathbb{Z}$,

$$C = M^e \pmod{n},$$

where C is the encrypted message.

Key Fact.

Given C, it is hard to find M such that

$$M^e = C,$$

where M is the original message.
Unless we have:

Decryption Key. There exists $d \in \mathbb{Z}$ such that

$$M = C^d.$$

This is private.

Corollary

If

$$a \in \mathbb{Z}/pq\mathbb{Z}, \qquad p, q \text{ primes}, \qquad \gcd(a, pq) = 1,$$

then

$$a^{(p-1)(q-1)} = 1.$$

Proof

Let $a \in \mathbb{Z}$. By Fermat's little theorem,

$$a^{(p-1)(q-1)} = (a^{q-1})^{p-1} \equiv 1 \pmod{p},$$
$$a^{(p-1)(q-1)} \equiv 1 \pmod{q}.$$

Hence p, q divide

$$a^{(p-1)(q-1)} - 1,$$

so

$$a^{(p-1)(q-1)} \equiv 1 \pmod{n}.$$

Claim.

If

$$de \equiv 1 \pmod{(p-1)(q-1)},$$

then

$$C = M^e \Rightarrow M = C^d.$$

Proof

$$C^d = (M^e)^d = M^{ed} = M^1 \qquad \text{in } \mathbb{Z}/n\mathbb{Z},$$

by the corollary.

Idea. It is almost impossible to find p, q, only given n, e.

$$(\text{you find } d \iff \text{you find } p, q).$$

Numerical Calculations

Suppose M is 100×100.

Question. How can we calculate $\det(M)$?

Answer. Compute

$$\det M \pmod{2, 3, 5, 7, 11, 13, \ldots}.$$

If $\det M$ is known in

$$\mathbb{Z}/2\mathbb{Z}, \quad \mathbb{Z}/3\mathbb{Z}, \quad \ldots,$$

then we would know $\det M$ in

$$\mathbb{Z}/(2 \cdot 3 \cdot 5 \cdots)\mathbb{Z}$$

by the Chinese Remainder Theorem (later).

If

$$\begin{aligned} \det M &\equiv a_1 \pmod{p_1}, \\ &\vdots \\ \det M &\equiv a_n \pmod{p_n}, \end{aligned}$$

then

$$\det M \equiv \text{something unique} \pmod{p_1 \cdots p_n}.$$

This is NOT $a_1 a_2 \cdots a_n$!!

In particular,

$$\mathbb{Z}/(p_1 \cdots p_n)\mathbb{Z} \cong \mathbb{Z}/p_1\mathbb{Z} \times \mathbb{Z}/p_2\mathbb{Z} \times \cdots \times \mathbb{Z}/p_n\mathbb{Z}.$$

Polynomial Rings

Definition

Given any ring R, the polynomial ring $R[x]$ in one indeterminate x is

$$R[x] = \left\{ a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \;\middle|\; a_n, \ldots, a_0 \in R \right\}.$$

$+, \times$ on $R[x]$ is trivial.

Proposition

$$(R[x], +, \times)$$

is a ring.

Proof

View $f(x) \in R[x]$ as a function

$$f : R \to R.$$

Issue. It is NOT injective if R is a finite ring; that is,

$$\underbrace{f = g}_{F(R)} \not\Rightarrow \underbrace{f(x) = g(x)}_{R[x]}.$$

For example,

$$x^p - x = 0 \quad \text{in } F(R)$$

if

$$R = \mathbb{Z}/p\mathbb{Z}$$

by Fermat's little theorem, but

$$x^p - x \neq 0 \quad \text{in } R[x].$$

Discussion — New Rings from Old Ones

Example

$$\mathbb{Z}, \qquad \mathbb{Z}/m\mathbb{Z}.$$

1. Cartesian Product.

$$R_1 \times R_2 = \{(x_1, x_2) \mid x_1, x_2 \in R_2\}.$$

The operations are

$$(x_1, y_1) + (x_2, y_2) := (x_1 + y_1,\ x_2 + y_2),$$

and

$$(x_1, y_1)(x_2, y_2) := (x_1 y_1,\ x_2 y_2).$$

Then

$$(R_1 \times R_2, +, \times)$$

is a ring, and

$$0_{R_1 \times R_2} = (0_{R_1}, 0_{R_2}).$$

2. Let X be any set. Let R be a ring. Define

$$R^X := \{f : X \to R\}.$$

Addition and multiplication are pointwise. Then R^X forms a ring, with

$$0_{R^X} = \text{constant function } \longmapsto 0_R,$$

and

$$1_{R^X} = \text{constant function } \longmapsto 1_R.$$

Example If $X = \{a, b\}$, then

$$(R^X, +, \times) \cong (R \times R, +, \times)$$

by

$$f \longmapsto (f(a), f(b)).$$

Remark If A, B are sets,

$$A^B := \{f : B \to A\}.$$

The reason for this notation is that we can identify A^B as an ordered pair of elements of A; the size of the ordered pair depends on the size of B:

$$\underbrace{A \times A \times \cdots \times A}_{\text{size of } B}.$$

3. Subrings

Proposition

Let R be a ring and $S \subset R$. If S is closed under

$$(+), \quad \times, \quad -,$$

and $1 \in S$, then S is a subring (itself a ring).

> **Remark.** The condition $1 \in S$ is unnecessary under a definition of rings that does not require a multiplicative identity.

Example Let $R = \mathbb{C}$. Then

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\},$$

the Gaussian integers.

Example Let $R = \mathbb{Z}$, and

$$S = \{\text{even integers}\}.$$

S is closed, but $1 \notin S$. By our definition, S is not a subring!!

Natural Map between $R[x]$ and R^R

$$\varphi : R[x] \to R^R, \qquad f(x) \longmapsto f.$$

Note that φ preserves $+, \times$:

$$\varphi(ab) = \varphi(a)\varphi(b),$$
$$\varphi(a + b) = \varphi(a) + \varphi(b).$$

Thus it is a homomorphism.

Rule. If φ is injective, then we can inherit a lot of good properties from R^R to $R[x]\ldots$ Is it?

Polynomials

Let R be a ring. $R[x]$ is the polynomial ring.

Definition

The degree of $f(x) \in R[x]$ is the largest exponent of x. Also,

$$\deg 0 = -\infty.$$

Proposition

If R is an integral domain and $f, g \in R[x]$, then

$$\deg(f(x)g(x)) = (\deg f(x))(\deg g(x)).$$

Corollary

If R is an integral domain, then so is $R[x]$; that is,

$$fg = 0 \Rightarrow f = 0 \text{ or } g = 0.$$

Proof

Using the proposition.

Assumption. Assume now that R is a field F.

Examples (fields).

$$\mathbb{Q}, \quad \mathbb{R}, \quad \mathbb{C}, \quad \mathbb{Z}/p\mathbb{Z}, \quad \mathbb{Q}_p, \quad \mathbb{Q}[\sqrt{2}], \quad \mathbb{Q}[i\sqrt{5}], \quad \mathbb{Q} + i\mathbb{Q}.$$

The last field is called the Gaussian rationals.

Note. All of these are field extensions of $\mathbb{Q}$ or $\mathbb{Z}/p\mathbb{Z}$.

Example. There is a multiplicative inverse but the law of composition is not commutative (a "non-commutative field"). Hamilton's quaternions are

$$\mathbb{H} = \left\{ \begin{pmatrix} w & z \\ -\overline{z} & \overline{w} \end{pmatrix} \;\middle|\; w, z \in \mathbb{C} \right\}.$$

This is a non-commutative field/skew-field (not a field).

Division with Remainder

Proposition

If $f(x), g(x) \in F[x]$ and $g(x) \neq 0$, then there are unique $q(x), r(x) \in F[x]$ such that

$$f(x) = q(x)g(x) + r(x), \qquad \deg r(x) < \deg g(x).$$

Proof

By induction on degree, or using the well-ordering principle, let

$$\Omega := \{f(x) - q(x)g(x) \mid q(x) \in F[x]\},$$

and

$$\deg(\Omega) := \{\deg r(x) \mid r(x) \in \Omega\} \cap (\mathbb{N} \cup \{-\infty\}).$$

Case 1. If $-\infty \in \deg(\Omega)$, then $0 \in \Omega$. Thus there is a $q(x)$ such that

$$f(x) - q(x)g(x) = 0,$$

so $r(x) = 0$.

Case 2. Suppose $\deg(\Omega) \subset \mathbb{N}$. By the well-ordering principle, there is $r(x) \in \Omega$ having least degree in $\deg(\Omega)$.

Claim.

$\deg r(x) < \deg g(x)$.
Suppose otherwise, and write

$$r(x) = a_n x^n + \cdots, \qquad g(x) = b_m x^m + \cdots, \qquad m \leq n.$$

Then

$$\widetilde{r}(x) := r(x) - a_n b_m^{-1} x^{n-m} g(x) \in \Omega,$$

where $b_m^{-1} \in F$ and $a_n b_m^{-1} x^{n-m} \in F[x]$.

Claim. (continued)

Then

$$\deg(\widetilde{r}) \leq n - 1 < \deg r(x),$$

a contradiction. Hence $m > n$, and $r(x)$ is a valid remainder.

Uniqueness. Suppose

$$f(x) = q(x)g(x) + r(x),$$
$$f(x) = \widetilde{q}(x)g(x) + \widetilde{r}(x),$$

where $\deg \widetilde{r}, \deg r < \deg g$. Then

$$g(x)(q(x) - \widetilde{q}(x)) = \widetilde{r}(x) - r(x).$$

Thus

$$\deg g(x) + \deg(q - \widetilde{q}) = \deg(\widetilde{r} - r) < \deg g,$$

so $q - \widetilde{q} = 0$. Therefore $q = \widetilde{q}$, and hence $r = \widetilde{r}$. □

Divisibility

Definition

$f \mid g$ if there is $q \in F[x]$ such that $g = fq$.

Definition

The gcd of f, g is the unique monic polynomial of largest degree that divides both f and g. ("Monic" means that it starts with x^n, with leading coefficient 1.)

Note. If f and g are monic, and $f \mid g$ and $g \mid f$, then $f = g$.

Theorem

There exist $a(x), b(x) \in F[x]$ such that the gcd $d(x)$ of f, g satisfies

$$d(x) = a(x)f(x) + b(x)g(x).$$

Therefore every common divisor of f, g divides $d(x)$.

Proof

Let

$$A := \{a(x)f(x) + b(x)g(x) \neq 0 \mid a, b \in F[x]\}.$$

Let $d(x)$ be a smallest monic polynomial in A (smallest in degree). By remainder division,

$$f(x) = d(x)q(x) + r(x), \qquad \deg r(x) < \deg d(x).$$

Either

$$r(x) = f(x) - d(x)q(x) \in A,$$

or $r(x) = 0$. The former contradicts the minimality of d, because $r(x)$ is smaller.

Proof (continued)

Thus $r(x) = 0$, so

$$d(x) \mid f(x).$$

Similarly, $d(x) \mid g(x)$. Hence $d(x)$ is a common divisor. It is greatest and unique because, if $\widetilde{d}(x)$ is any common divisor, then from

$$d(x) = a(x)f(x) + b(x)g(x)$$

we obtain

$$\widetilde{d}(x) \mid d(x), \qquad \deg \widetilde{d}(x) \leq \deg d(x).$$

If there is another gcd $\widetilde{d}$, then

$$\widetilde{d} \mid d, \qquad d \mid \widetilde{d}.$$

Since both are monic, $d = \widetilde{d}$. $\square$

Definition

A polynomial $f(x) \in F[x]$ is irreducible if

$$f(x) = p(x)q(x) \quad \Longrightarrow \quad \deg p(x) = 0 \text{ or } \deg q(x) = 0.$$

Correction. Polynomials of degree 0, or degree $-\infty$, are not irreducible.

Remark Whether $f(x)$ is irreducible depends on F.

Examples.

$$x^2 + 1 \quad \text{is irreducible in } \mathbb{R}[x],$$

but

$$x^2 + 1 = (x - i)(x + i) \quad \text{in } \mathbb{C}[x].$$

Also,

$$x^2 - 2 \quad \text{is irreducible in } \mathbb{Q}[x],$$

but

$$x^2 - 2 = (x - \sqrt{2})(x + \sqrt{2}) \quad \text{in } \mathbb{R}[x].$$

Theorem

There are infinitely many irreducible polynomials in $F[x]$, for every field F.

Proof

Assume $P_1, \ldots, P_N$ are all irreducible polynomials. Set

$$f(x) = P_1 P_2 \cdots P_N + 1.$$

The polynomial $f(x)$ has to be irreducible; otherwise one of the P_j's divides 1, a contradiction. □

Gauss's Lemma.

If $f(x)$ is irreducible, then

$$f(x) \mid g_1 g_2 \quad \Longrightarrow \quad f \mid g_1 \text{ or } f \mid g_2.$$

Proof

If $f \mid g_1$, we are done. Suppose $f \nmid g_1$. Then $(f, g_1) = 1$, since f is irreducible. Hence

$$a(x)f(x) + g_1(x)b(x) = 1.$$

Multiplying by $g_2(x)$,

$$a(x)f(x)g_2(x) + g_1(x)g_2(x)b(x) = g_2(x).$$

The terms on the left are divisible by $f(x)$, so

$$f(x) \mid g_2(x).$$

□

Consequence. If $f \mid g_1 \cdots g_n$ and f is irreducible, then $f \mid g_j$ for some j.

Unique Factorization Theorem

Theorem

If $f(x) \in F[x]$, then it can be written uniquely as

$$f(x) = \lambda\, g_1^{e_1}(x) g_2^{e_2}(x) \cdots g_t^{e_t}(x),$$

where $\lambda \in F^\times$, the polynomials $g_1, \ldots, g_t$ are irreducible, monic, and distinct, and

$$e_1, \ldots, e_t \geq 1.$$

Definition

A ring endowed with a size/Euclidean/gauge function

$$|\cdot| : R \longrightarrow \mathbb{N}$$

such that

$$|fg| \geq |f| \text{ and } |g|, \qquad \forall f, g \neq 0,$$

and such that for all f, g there exist q, r with

$$f = gq + r, \qquad |r| < |g|,$$

is a Euclidean domain.

Roots of Polynomials

Definition

If $f(x) \in F[x]$ and $r \in F$, then r is a root of $f(x)$ if

$$f(r) = 0.$$

Theorem

r is a root of $f(x)$ if and only if

$$x - r \mid f(x).$$

Proof

For ($\Rightarrow$), divide $f(x)$ by $x - r$.

Proof (continued)

The remainder is 0, so $(x - r) \mid f(x)$. The reverse implication is trivial. $\square$

Corollary

If $f(x) \in F[x] \setminus \{0\}$, then f has at most $n = \deg f$ roots.

Proof

Let $r_1, \ldots, r_k$ be roots of $f(x)$. Then

$$(x - r_1) \cdots (x - r_k) g(x) = f(x).$$

Comparing degrees gives

$$k \leq \deg f(x).$$

□

Corollary

The map

$$\varphi : F[x] \longrightarrow F^F$$

is one-to-one if F is infinite.

Proof

If $\varphi(f(x)) = \varphi(g(x))$, then

$$f(t) = g(t), \qquad \forall t \in F.$$

Thus $(f - g)(x)$ has infinitely many roots. By the previous corollary, $(f - g)(x) = 0$, and hence $f(x) = g(x)$. □

Remark If F is finite, φ must not be bijective. Indeed,

$$|F[x]| = |\mathbb{N}|, \qquad |F^F| = |F|^{|F|},$$

so φ must not be injective.

Remark A significant counterexample to injectivity is

$$x(x-1) \cdots (x - (p-1)) = f(x).$$

All elements of $\mathbb{Z}/p\mathbb{Z}$ are roots of it. Thus

$$f(x) \neq 0, \qquad \text{but} \qquad f = 0 \text{ in } F^F.$$

Moreover, Corollary

$$x(x-1) \cdots (x - (p-1)) = x^p - x.$$

Proof

By Fermat's little theorem,

$$x^p - x = 0, \qquad \forall x \in \mathbb{Z}/p\mathbb{Z}.$$

Thus, by the unique factorization theorem, the two polynomials must be equal. □

Recall. The following features of $F[x]$ are independent of F:

$$\left.\begin{array}{c}\text{division with remainder,}\\ \text{greatest common divisor algorithm,}\\ \text{Gauss's lemma,}\\ \text{unique factorization}\end{array}\right\} \longrightarrow \text{independent of } F.$$

Question. What are the irreducible polynomials in $F[x]$? How do we determine/factorize them?

Answer. It depends on the field F.

$F = \mathbb{C}$. (1) All monic irreducible polynomials are of the form

$$x - a, \qquad a \in \mathbb{C}.$$

Fundamental Theorem of Algebra

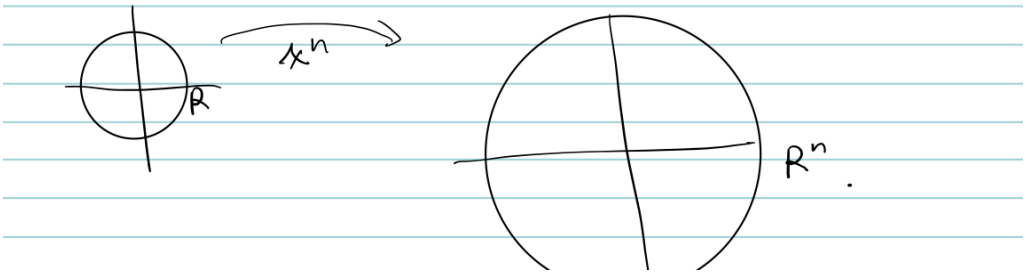
Theorem

If $f(x) \in \mathbb{C}[x]$ and $\deg f(x) \geq 1$, then $f(x)$ has a root in $\mathbb{C}$.

Idea of Proof

(Not using Liouville's theorem.) Write

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0.$$



The circle of radius R and its image under $z \mapsto z^n$, a circle of radius R^n.

As $R \to \infty$, the term other than x^n is little, so there is $C > 0$ such that

$$\left|a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_0\right| < CR^{n-1}.$$

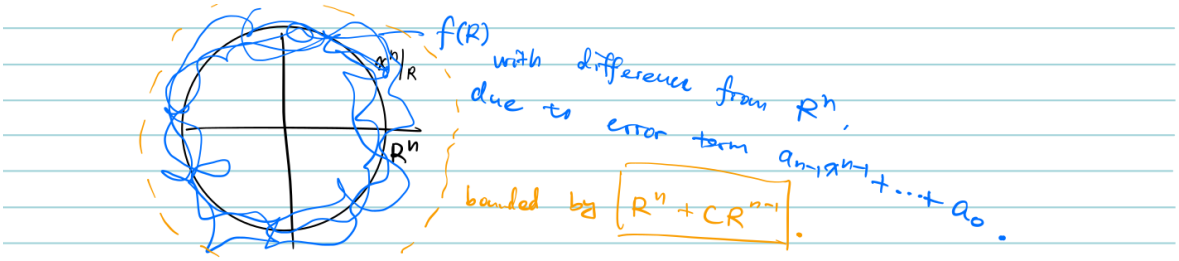
Then

$$|f(R)| = |R^n + \cdots| \leq |R^n| + |\cdots| < R^n + CR^{n-1}.$$

Moreover. $f(R)$ is continuous. The $f(R)$-curve differs from the R^n-circle by the error term

$$a_{n-1}x^{n-1} + \cdots + a_0,$$

and is bounded by $R^n + CR^{n-1}$.



For large R, the image curve $f(R)$ is a perturbation of the radius-R^n circle.

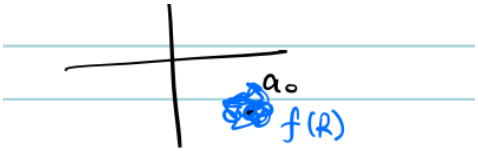
Now if $R \to 0$ (real small), then

$$f(R) \longrightarrow a_0 \neq 0,$$

while

$$x^n + \cdots + a_1 x$$

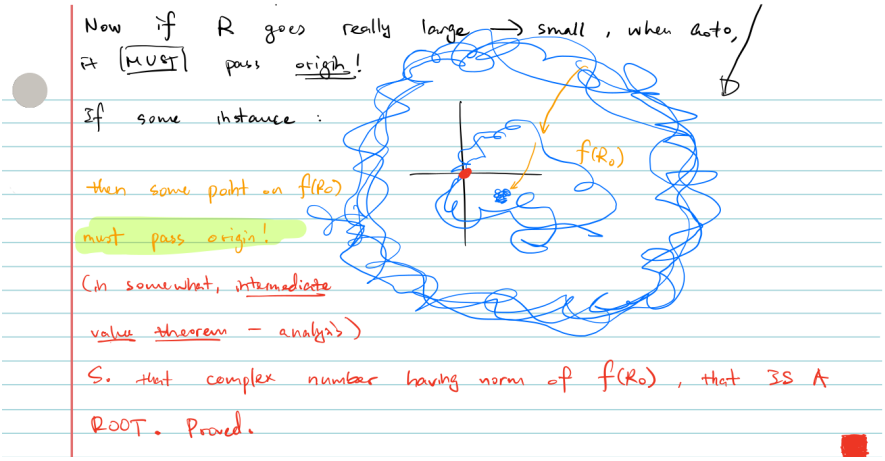
becomes an error term tending to 0. Thus the term becomes an error around a_0.



For small R, the image $f(R)$ lies near a_0.

Now if R goes really large $\longrightarrow$ small, when $a_0 \neq 0$, it *must* pass the origin.

If, at some instance R_0, some point on $f(R_0)$ passes the origin (in a somewhat intermediate-value-theorem analysis), then that complex number "having norm of $f(R_0)$" is a root. Proof. ■



As R decreases, the continuously varying image curve passes through the origin at an intermediate radius R_0.

Fundamental Theorem of Algebra

If $f \in \mathbb{C}[x]$ is monic, then

$$f(x) = (x - r_1)^{n_1}(x - r_2)^{n_2} \cdots (x - r_k)^{n_k}$$

uniquely, where $r_1, \ldots, r_k$ are the roots of f.

Question. How do we calculate $r_1, r_2, \ldots, r_k$?

Answer. Newton–Raphson method.

$F = \mathbb{R}$. If $f(x) \in \mathbb{R}[x] \subset \mathbb{C}[x]$, then in $\mathbb{C}[x]$,

$$f(x) = (x - r_1)^{n_1} \cdots (x - r_k)^{n_k}, \qquad r_1, \ldots, r_k \text{ roots in } \mathbb{C}.$$

If r is a root of f, then

$$f(\overline{r}) = \overline{f(r)} = \overline{0} = 0.$$

Theorem

The roots of $f(x) \in \mathbb{R}[x]$ are either real or occur in complex conjugate pairs. □

Corollary

The monic irreducible polynomials in $\mathbb{R}[x]$ are of two kinds:

1. $$x - r, \qquad r \in \mathbb{R};$$

2. $$\begin{aligned}(x - (a + bi))(x - (a - bi)) &= x^2 - 2ax + (a^2 + b^2) \\ &= x^2 + rx + t, \qquad \Delta = r^2 - 4t < 0.\end{aligned}$$

$F = \mathbb{Q}$.

Eisenstein's Criterion

Suppose

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0, \qquad a_j \in \mathbb{Z}.$$

Suppose there is a prime p such that

$$p \mid a_j \quad \text{and} \quad p^2 \nmid a_0.$$

Then $f(x)$ is irreducible in $\mathbb{Q}[x]$.

Lemma.

For $f(x) \in \mathbb{Z}[x]$, if there are $g, h \in \mathbb{Q}[x]$ such that

$$g(x)h(x) = f(x),$$

then there is a factorization in $\mathbb{Z}[x]$.

Proof

There is $N = N_1 N_2 \in \mathbb{Z}$ such that

$$Nf(x) = (N_1 g(x))(N_2 h(x)), \qquad N_1 g(x), N_2 h(x) \in \mathbb{Z}[x].$$

Consider a prime $q \mid N$, and reduce the coefficients modulo $\mathbb{Z}/q\mathbb{Z}$. Then

$$0 = \overline{N_1 g(x)}\,\overline{N_2 h(x)} \quad \text{in } (\mathbb{Z}/q\mathbb{Z})[x].$$

Now $(\mathbb{Z}/q\mathbb{Z})[x]$ is an integral domain (trivial), so

$$\overline{N_1 g(x)} = 0 \quad \text{or} \quad \overline{N_2 h(x)} = 0$$

as a polynomial. Without loss of generality, suppose $\overline{N_1 g(x)} = 0$. Then $N_1 g(x)$ has all its coefficients as multiples of q. Hence

$$\frac{N_1 g(x)}{q} \in \mathbb{Z}[x],$$

and

$$\frac{N}{q} f(x) = \frac{N_1}{q} g(x)\, N_2 h(x).$$

Keep doing this for every prime factor of N. Eventually,

$$\frac{N}{N} f(x) = \widetilde{g}(x)\widetilde{h}(x), \qquad \widetilde{g}, \widetilde{h} \in \mathbb{Z}[x].$$

□

Proof of Eisenstein's Criterion

Suppose, for a contradiction,

$$f(x) = h(x)g(x), \qquad g(x), h(x) \in \mathbb{Q}[x].$$

By the lemma, we may assume $g(x), h(x) \in \mathbb{Z}[x]$ and monic.

Proof of Eisenstein's Criterion (continued)

Reduce this modulo $p\mathbb{Z}[x]$, with p as in the assumption. Then

$$\overline{f(x)} = x^n = \overline{g(x)}\,\overline{h(x)}.$$

Thus there is $k \geq 1$ such that

$$\overline{g(x)} = x^k, \qquad \overline{h(x)} = x^{n-k}.$$

Hence

$$g(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_1 x + b_0,$$
$$h(x) = x^{n-k} + c_{n-k-1}x^{n-k-1} + \cdots + c_0,$$

where

$$b_j, c_j \in \mathbb{Z}, \qquad p \mid b_j, \quad p \mid c_j.$$

It follows that

$$a_0 = b_0 c_0 = p \cdot p \cdot (\ ) = p^2(\ ),$$

so

$$p^2 \mid a_0,$$

a contradiction. □

$F = \mathbb{Z}/p\mathbb{Z}$.

Theorem

For $f(x) \in (\mathbb{Z}/p\mathbb{Z})[x]$,

$$\#\{\text{roots of } f(x)\} = \deg(\gcd(x^p - x, f(x))).$$

Proof

Recall that, by Fermat's little theorem and unique factorization,

$$x^p - x = \prod_{j \in \mathbb{Z}/p\mathbb{Z}} (x - j).$$

It follows from the fact that

$$j \text{ is a root of } f(x) \quad \iff \quad (x - j) \mid f(x).$$

□

Congruences in $F[x]$

Definition

We say

$$f(x) \equiv g(x) \pmod{m(x)}$$

if

$$m(x) \mid f(x) - g(x).$$

Theorem

Congruence is an equivalence relation, and it is compatible with $+$ and $\times$.

Definition

The collection of congruence classes modulo $m(x)$ is denoted by

$$F[x]/m(x) := \{a_0 + a_1x + \cdots + a_{n-1}x^{n-1} \mid a_i \in F\}.$$

Proposition

$F[x]/m(x)$ is a ring.

Proof

$$[g_1(x)] + [g_2(x)] = [g_1(x)g_2(x)],$$
$$[g_1(x)]\,[g_2(x)] = [g_1(x)g_2(x)].$$

□

Remark

$$F^n \cong F[x]/f(x)$$

as vector spaces, from compatibility of $+$ and scalar multiplication.

Proposition

$$F[x]/p(x) \text{ is a field} \iff p(x) \text{ is irreducible.}$$

Proof

For a nonzero class represented by $f(x)$, Bézout gives

$$\underbrace{a(x)p(x)}_{0} + b(x)f(x) = 1.$$

□

Example.

$$\mathbb{R}[x]/(x^2+1) = \{[a+bx] \mid a, b \in \mathbb{R}\}.$$

Multiplication is

$$[a+bx][c+dx] = [ac + bdx^2 + (ad+bc)x].$$

Since

$$x^2 + 1 = 0 \implies x^2 = -1,$$

we obtain

$$[a+bx][c+dx] = [ac - bd + (ad+bc)x] \cong \mathbb{C}.$$

Under this isomorphism,

$$\sqrt{-1} \longmapsto [x] \in \mathbb{R}[x]/(x^2+1).$$

Example.

$$\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z} = \{0, 1\}.$$

Then

$$\mathbb{F}_2[x]/(x^2 + x + 1)$$

is a field of 4 elements. Its addition and multiplication tables are

$+$	0	1	x	$x+1$
0	0	1	x	$x+1$
1	1	0	$x+1$	x
x	x	$x+1$	0	1
$x+1$	$x+1$	x	1	0

$\times$	0	1	x	$x+1$
0	0	0	0	0
1	0	1	x	$x+1$
x	0	x	$x+1$	1
$x+1$	0	$x+1$	1	x

(a field).

Example. Since

$$[x^3] = 2,$$

we may write $[x] = \sqrt[3]{2}$. Thus

$$\mathbb{Q}[x]/(x^3 - 2) = \mathbb{Q}[\sqrt[3]{2}],$$

generated over $\mathbb{Q}$ by $\sqrt[3]{2}$.

Recall. If

$$K = F[x]/p(x),$$

then K is a field if and only if $p(x)$ is irreducible.

Theorem

Let F be a field, and let $p(x)$ be an irreducible polynomial in $F[x]$. Then there is a field K which contains F, as well as a root $\alpha \in \mathbb{C}$ of $p(x)$. By the Fundamental Theorem of Algebra,

$$F[\alpha] = K.$$

Proof

Let

$$K = F[x]/p(x).$$

The set K is a field by the proposition proved last time. We have

$$F \subset K, \qquad F = \{[a] \mid a \in F\} \subset K,$$

where the elements on the right are degree-0 polynomials.
Let

$$\alpha = [x].$$

Then α is a root of $p(x)$. Indeed, if

$$p(x) = a_n x^n + \cdots + a_1 x + a_0,$$

then

$$p(\alpha) = a_n[x]^n + \cdots + a_1[x] + a_0 = [p(x)] = [0].$$

Notation. From now on,

$$\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p \quad \text{or} \quad \mathbb{Z}_p.$$

More General Rings

Example. Matrices:

$$M_n(R) = M_{n\times n}(R),$$

where R is any commutative ring.

Example. On $\mathcal{P}(X)$, define

$$A + B = A \cup B \setminus (A \cap B) = A \triangle B, \qquad A \times B = A \cap B.$$

Then

$$0 = \varnothing, \qquad 1 = X.$$

Example.

$$R^X = \mathcal{F}(X, R) = \{f : X \longrightarrow R\}.$$

Example. $\mathbb{Z}[x]$ fails to be a Euclidean domain because not every integer has an inverse. Thus the proof fails starting from integer division.

Example.

$$F[x, y], \qquad \text{or} \qquad F[x_1, x_2, \ldots, x_n].$$

Ideals

Definition

For a commutative ring R, an ideal I satisfies:

1. $I \subset R$ and $I \neq \varnothing$;
2. $a, b \in I \Longrightarrow a + b \in I$;
3. for every $\lambda \in R$ and $a \in I$,

$$\lambda a \in I.$$

Side note. The $a + b$ condition suffices because $-1 \in R$, so $-a \in R$.

Example.

$$\{0\},\ R \subset R$$

are the trivial ideals of any commutative ring R.

Examples of nontrivial ideals.

$$2\mathbb{Z} = \{\text{even integers}\}.$$

Also,

$$I := \{\text{polynomials with zero constant term}\} \subset F[x].$$

If $R = F[x]$ and $p(x) \in F[x]$ is fixed, then

$$I := \{p(x)f(x)\}$$

is an ideal. Finally,

$$n\mathbb{Z} \subset \mathbb{Z}$$

is an ideal.

Definition

Let R be a ring. Define

$$(a) = Ra := \{\lambda a \mid \lambda \in R\}$$

to be the ideal generated by a. An ideal generated by one element in this way is called a principal ideal.

Theorem

Let I be an ideal in $\mathbb{Z}$ or $F[x]$, where F is a field. Then I is principal.

Proof

Let $a \in I \setminus \{0\}$ be an element of smallest size/degree/absolute value in I.

Proof (continued)

If $b \in I \setminus \{0\}$, then

$$b = aq + r, \qquad \text{size}(r) < \text{size}(a).$$

Since $r \in I$ by closure, a nonzero r would contradict the minimality of a. Thus

$$b = aq, \qquad I = aR,$$

so I is principal. $\square$

Recall. We proved this when proving that $\gcd(a, b)$ is the least element in

$$Ra + Rb.$$

Note. If R is noncommutative, there are left-sided ideals, right-sided ideals, and two-sided ideals.

Example. Let

$$R = F[x, y], \qquad I = (x, y) = Rx + Ry.$$

The ideal I is not principal. If $I = (f)$, then

$$f \mid x, \qquad f \mid y,$$

so f is a nonzero constant in F. Yet

$$I \cap F = \{0\},$$

since

$$I = \{f(x,y) \mid f(0,0) = 0\}.$$

Indeed, if $f(0,0) = 0$, then

$$f = \sum_{i+j>0} a_{ij} x^i y^j,$$

because setting $x = y = 0$ gives $a_{00} = 0$. Thus there are g, h such that

$$f = gx + hy.$$

Therefore $I \cap F = \{0\}$, so I cannot be (f). □

Example. Consider $(2, x)$ in $\mathbb{Z}[x]$:

$$I = (2,x) = 2\mathbb{Z}[x] + x\mathbb{Z}[x] = \{f(x) \mid \overline{f(0)} = 0 \in \mathbb{Z}/2\mathbb{Z}\}.$$

The ideal I is not principal, by the same argument. If $(f) = I$, then

$$f = \pm 1,$$

but odd numbers are not in I.

Example. In $\mathbb{Z}[\sqrt{-5}]$, let

$$I = \{a + b\sqrt{-5} \mid 3 \mid a \text{ and } 3 \mid b\} = (3, 1 - \sqrt{-5}).$$

It is an ideal (exercise). Both

$$3 \in I, \qquad 1 - \sqrt{-5} \in I.$$

The note concludes that I is nonprincipal. The accompanying norm calculation is

$$\|1 - \sqrt{-5}\|^2 = 6, \qquad \|3\|^2 = 9.$$

The possible norm of a common divisor is marked as 3 or 1; the 3-case is crossed out, leaving only the common divisors ± 1.

Question. Why ideals?

Answer.

$$\text{principal ideal domain} \Longrightarrow \text{unique factorization domain}; \qquad \text{quotient ring } R/I.$$

Consequences

$$R/I := \{\lambda + I = [\lambda] \mid \lambda \in R\},$$

with

$$[\lambda_1] = [\lambda_2] \iff \lambda_2 - \lambda_1 \in I.$$

Definition

A ring whose ideals are all principal is called a principal ideal domain.

Counterexamples.

$$\mathbb{Z}[x] \supset (2, x), \qquad F[x, y] \supset (x, y), \qquad \mathbb{Z}[\sqrt{-5}] \supset (3, 1 - \sqrt{-5}).$$

Examples.

$$F[x], \qquad \mathbb{Z}.$$

Other example of a principal ideal domain. The Gaussian integers are

$$\mathbb{Z} + i\mathbb{Z} = R.$$

Definition

Consider the size function

$$s : R \longrightarrow \mathbb{N}, \qquad a + bi \longmapsto a^2 + b^2.$$

Proposition

It preserves multiplication and satisfies positive-definiteness.

Theorem

For

$$z_1 = x_1 + iy_1, \qquad z_2 = x_2 + iy_2 \neq 0$$

in $\mathbb{Z}[i]$, there exist $q, r \in R$, not necessarily uniquely, such that

$$z_1 = qz_2 + r, \qquad s(r) < s(z_2).$$

Proof

By division in $\mathbb{C}$, $z_2^{-1} \in \mathbb{C}$, and

$$z_1 z_2^{-1} = a + bi \in \mathbb{C}.$$

Round $a, b \in \mathbb{R}$ to the nearest integers $\widetilde{a}, \widetilde{b}$. Then

$$q := \widetilde{a} + \widetilde{b}\, i, \qquad z_1 z_2^{-1} = q + \varepsilon,$$

where

$$\varepsilon = \varepsilon_1 + \varepsilon_2 i, \qquad \varepsilon_1 = a - \widetilde{a}, \quad \varepsilon_2 = b - \widetilde{b}, \qquad |\varepsilon_1|, |\varepsilon_2| \leq \frac{1}{2}.$$

Thus

$$z_1 = qz_2 + \underbrace{\varepsilon z_2}_{r},$$

and

$$s(r) = s(\varepsilon)s(z_2) \leq \left(\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right) s(z_2) = \frac{1}{2}s(z_2).$$

Note that $r \in \mathbb{Z}[i]$, since

$$z_1 - qz_2 = \varepsilon z_2.$$

□

Theorem

Euclidean domain $\implies$ principal ideal domain.

The proof is the same as for polynomials. Thus

$$\boxed{\mathbb{Z}[i] \text{ is a principal ideal domain.}}$$

Theorem

If p is a prime of the form $4k+1$, then

$$p = a^2 + b^2$$

for some $a, b \in \mathbb{Z}$.

Note. It is an if-and-only-if statement. The side calculation reads

$$p = a^2 + b^2 \quad \implies \quad a^2 \equiv 0 \text{ or } 1 \pmod 4, \quad b^2 \equiv 0 \text{ or } 1 \pmod 4,$$

and hence

$$p \equiv 0 \text{ or } 1 \text{ or } 2 \pmod 4;$$

since p is odd, $p \equiv 1 \pmod 4$.

Lemma.

If $p = 1 + 4k$, then there is an integer t with

$$-\frac{p}{2} \leq t \leq \frac{p+1}{2}$$

such that

$$p \mid t^2 + 1.$$

Proof

In $\mathbb{F}_p[x]$,

$$x^{p-1} - 1 = x^{4k} - 1 = (x^{2k} - 1)(x^{2k} + 1).$$

The polynomial $x^{2k} + 1$ splits into linear factors by Fermat's little theorem. Let $r \in \mathbb{Z}/p\mathbb{Z}$ be a root of it. Then

$$t := r^k \in \mathbb{Z}/p\mathbb{Z}$$

works. Let t be the representative of $\mathbb{Z}/p\mathbb{Z}$ satisfying

$$\frac{-p+1}{2} \leq t \leq \frac{p+1}{2}.$$

□

Proof

Consider

$$t + i \in \mathbb{Z}[i].$$

By the preceding lemma,

$$s(t+i) = t^2 + 1 = mp.$$

Proof (continued)

Moreover,

$$t^2 + 1 = mp \le \frac{p^2}{4} < p^2 \qquad \left(\text{since } \frac{-p+1}{2} \le t \le \frac{p+1}{2}\right) \qquad \Longrightarrow \qquad p \nmid m.$$

Consider

$$I = (p, t+i) \qquad \text{(an ideal of } \mathbb{Z}[i]).$$

Since $\mathbb{Z}[i]$ is a principal ideal domain, there exists $a + bi \in \mathbb{Z}[i]$ such that

$$I = (a + bi).$$

Thus $a + bi$ divides $t + i$ and p in $R = \mathbb{Z}[i]$.
Then there exist $w, u \in \mathbb{Z}[i]$ such that

$$t + i = (a + bi)w, \qquad t - i = (a - bi)\overline{w} \qquad \text{(not hard to verify)}.$$

Hence

$$t^2 + 1 = (a^2 + b^2)w\overline{w} = mp.$$

Also,

$$p = (a + bi)u, \qquad p = (a - bi)\overline{u} \qquad \Longrightarrow \qquad p^2 = (a^2 + b^2)u\overline{u}.$$

Therefore

$$a^2 + b^2 = 1, p, \text{ or } p^2.$$

Since

$$mp = (a^2 + b^2)\underbrace{w\overline{w}}_{\in \mathbb{Z}},$$

we have $a^2 + b^2 \neq p^2$, so $a^2 + b^2 = 1$ or p.
For every $pz_1 + (t+i)z_2 \in \mathbb{Z}[i]$,

$$\begin{aligned} s(pz_1 + (t+i)z_2) &\equiv s\big((t+i)z_2\big) \\ &= s(t+i)s(z_2) \\ &= (t^2+1)s(z_2) \equiv 0 \pmod{p}, \end{aligned}$$

where $t^2 + 1$ is a multiple of p. Therefore, since $a + bi \in I$,

$$s(a + bi) = a^2 + b^2 \equiv 0 \pmod{p}.$$

Consequently,

$$a^2 + b^2 = p.$$

□

Congruences

Definition

Given an ideal $I \subset R$ and $a, b \in R$, we can define

$$a \equiv b \pmod{I} \qquad \iff \qquad a - b \in I.$$

Lemma.

The relation $\equiv$ is an equivalence relation.

Proof

This follows from the properties of an ideal.

Proposition

If $a \equiv a'$ and $b \equiv b' \pmod{I}$, then

$$ab \equiv a'b', \qquad a + b \equiv a' + b' \pmod{I}.$$

Definition

The *quotient of the ring R by the ideal I*, denoted R/I, is the set of equivalence classes modulo I.

Definition (continued)

$$R/I = \{a + I \mid a \in R\}.$$

Proposition

R/I is a ring with

$$(a + I) + (b + I) = (a + b) + I,$$
$$(a + I)(b + I) = ab + I.$$

Example If $R = \mathbb{Z}$ and $I = n\mathbb{Z}$, then

$$R/I = \mathbb{Z}/n\mathbb{Z}.$$

Side note (Chinese Remainder Theorem). If $(n, m) = 1$, then

$$\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}.$$

Choose $r, s \in \mathbb{Z}$ such that $rn + sm = 1$. An isomorphism is given by

$$(a, b) \longmapsto a(1 - rn) + b(1 - sm).$$

Example If $R = \mathbb{F}[x]$ and $I = (p(x))$, then

$$R/I = \mathbb{F}[x]/p(x).$$

Example Let $R = \mathbb{Z}[x]$ and $I = (2, x)$. Then

$$R/I = \{0 + I, 1 + I\} \cong \mathbb{Z}/2\mathbb{Z}.$$

Example Let $R = \mathbb{Z}[x]$ and $I = (2x)$. Then

$$R/I = \{a_0 + b_1 x + \cdots + b_n x^n \mid a_0 \in \mathbb{Z},\ b_i \in \{0, 1\}\}.$$

Exercise. Describe R/I explicitly and show that it is a ring:

$$R/I = \mathbb{Z} \oplus x((\mathbb{Z}/2\mathbb{Z})[x]).$$

Example Let

$$R = \mathbb{Z}[x, y], \qquad I = (5x, x^2 + y^2 - 3).$$

Homomorphisms

Definition

A homomorphism between two rings is a map

$$\varphi : R_1 \longrightarrow R_2$$

such that

$$\varphi(a + b) = \varphi(a) + \varphi(b), \qquad \varphi(ab) = \varphi(a)\varphi(b), \qquad \varphi(1_{R_1}) = 1_{R_2}.$$

Example There is no homomorphism

$$\varphi : \mathbb{Z}/n\mathbb{Z} \longrightarrow \mathbb{Z}.$$

Indeed, if such a φ existed, then $\varphi([1]) = 1$ by definition, and hence

$$\varphi([n]) = \varphi([1] + \cdots + [1]) = 1 + \cdots + 1 = n.$$

Yet

$$\varphi([n]) = \varphi([0]) = 0 \neq n.$$

Example Given $a \in \mathbb{F}$, the evaluation homomorphism (also denoted EV_a) is

$$\varphi_a : \mathbb{F}[x] \longrightarrow \mathbb{F}, \qquad f(x) \longmapsto f(a).$$

This is a ring homomorphism. This is also true for

$$R[x] \longrightarrow R, \qquad a \in R,$$

for a general ring R.

Definition

The kernel of a homomorphism $\varphi : R_1 \to R_2$ is

$$\ker \varphi := \{a \in R_1 \mid \varphi(a) = 0_{R_2}\}.$$

Proposition

Let $\varphi : R_1 \to R_2$ be a homomorphism. Then $\operatorname{Im}\varphi$ is a subring of R_2, and $\ker\varphi$ is an ideal of R_1.

Proposition (continued)

Given an ideal $I \subset R$, consider

$$\varphi : R \longrightarrow R/I, \qquad a \longmapsto a + I.$$

It is a homomorphism, and

$$\begin{aligned}\ker\varphi &= \{a \in R \mid \varphi(a) = 0 + I\} \\ &= I,\end{aligned}$$

since

$$a + I = 0 + I \qquad \Longleftrightarrow \qquad a - 0 = a \in I.$$

First Isomorphism Theorem

Let R be a ring and $I \subset R$ an ideal. If

$$\varphi : R \longrightarrow R'$$

has $\ker\varphi = I$, then

$$R/I \xrightarrow{\sim} \operatorname{Im}(\varphi)$$

is an isomorphism.

Proof

Define the homomorphism

$$\overline{\varphi} : R/I \longrightarrow S, \qquad a + I \longmapsto \varphi(a).$$

Verification is trivial. □

Recall: notation.

$$\begin{aligned} f : A \hookrightarrow B &\quad \text{means injective,} \\ f : A \twoheadrightarrow B &\quad \text{means surjective,} \\ f : A \xrightarrow{\sim} B &\quad \text{means bijective.}\end{aligned}$$

Application

Example Let

$$R = \mathbb{Z}, \qquad I = (n) = n\mathbb{Z}.$$

Then $R/I = \mathbb{Z}/n\mathbb{Z}$, as follows from the Euclidean algorithm.

Example Let

$$R = \mathbb{F}[x], \qquad I = (p(x)).$$

Then $R/I = \mathbb{F}[x]/p(x)$, as follows from the Euclidean algorithm.

Example (What if?) Let

$$R = \mathbb{F}[x_1, x_2, \ldots, x_n], \qquad I = (f_1, \ldots, f_m).$$

Then

$$R/I = \{g(x_1, \ldots, x_n) + I \mid g \in \mathbb{F}[x_1, \ldots, x_n]\}.$$

Given $g_1 + I$ and $g_2 + I$, are they equal?

General strategy for understanding R/I. Find

$$\varphi : R \longrightarrow S \qquad \text{such that} \qquad \ker \varphi = I.$$

Then

$$R/I \cong \mathrm{Im}(\varphi) \subset S.$$

Example Let

$$R = \mathbb{R}[x], \qquad I = (x^2 + 1),$$

and define

$$\varphi : R \longrightarrow S = \mathbb{C}, \qquad f(x) \longmapsto f(i).$$

Here

$$\gcd(x^2 + 1, f(x)) = 1 \text{ or } x^2 + 1, \qquad f(i) = 0 \iff x^2 + 1 \mid f(x).$$

Thus φ induces an isomorphism

$$\mathbb{R}[x]/(x^2 + 1) \xrightarrow{\sim} \mathrm{Im}\, \varphi.$$

Example Let

$$R = \mathbb{Z}[x], \qquad I = (2, x),$$

and define

$$\varphi : \mathbb{Z}[x] \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}, \qquad f(x) \longmapsto f(0) \pmod 2.$$

We can verify that φ is a homomorphism and

$$\ker \varphi = (2, x) = I.$$

Consequently,

$$\mathbb{Z}[x]/(2, x) \cong \mathbb{Z}/2\mathbb{Z} = \mathrm{Im}\, \varphi.$$

Example Let

$$R = \mathbb{Z}[x, y], \qquad I = \{f(x, y) \mid f(1, 2) = 0 \pmod p\}.$$

Define

$$\varphi : \mathbb{Z}[x, y] \twoheadrightarrow \mathbb{Z}/p\mathbb{Z}, \qquad f(x, y) \longmapsto f(1, 2) \pmod p.$$

Then

$$\ker \varphi = I \qquad \Longrightarrow \qquad R/I \cong \mathbb{Z}/p\mathbb{Z}.$$

Example Let

$$R = \mathbb{F}[x_1, x_2, \ldots, x_n], \qquad I = (f_1, \ldots, f_m).$$

Prime Ideals and Maximal Ideals

Definition

An ideal $I \neq R$ is *maximal* if, for every ideal J,

$$J \supsetneq I \qquad \Longrightarrow \qquad J = R.$$

Definition

An ideal $I \subset R$ is *prime* if

$$xy \in I \qquad \Longrightarrow \qquad x \in I \text{ or } y \in I, \qquad \forall x, y \in R.$$

This is a generalization of Gauss's lemma.

Example For $R = \mathbb{Z}$, I is a prime ideal if and only if

$$I = p\mathbb{Z}$$

for p prime.

Theorem

An ideal $I \subset R$ is maximal if and only if R/I is a field.

Proof

($\Rightarrow$) Suppose I is maximal. Let

$$a + I \in (R/I)^*.$$

We want to show that $a + I$ has a multiplicative inverse. Since

$$a + I \neq 0 + I,$$

we have $a \notin I$. Consider

$$J = Ra + I,$$

which is also an ideal, and $J \supsetneq I$. Therefore $J = R$, so there exist $x \in R$ and $y \in I$ such that

$$xa + y = 1.$$

Thus

$$xa + I = 1 + I \qquad \Longrightarrow \qquad (x + I) = (a + I)^{-1}.$$

Hence R/I is a field.

($\Leftarrow$) Suppose R/I is a field. Let $J \supsetneq I$; we want to show $J = R$. Choose $a \in J \setminus I$. Then

$$\exists (a + I)^{-1} = a' + I \in R/I,$$

so $aa' \in J$ since J is an ideal. Also

$$aa' + I = 1 + I, \qquad I \subsetneq J,$$

and hence

$$aa' - 1 \in I \subset J.$$

Therefore $1 \in J$, and

$$J = R1 = R.$$

□

Theorem

An ideal $I \subset R$ is a prime ideal if and only if R/I is an integral domain.

Proof

($\Rightarrow$) If $x + I, y + I \in R/I$ and

$$(x + I)(y + I) = 0,$$

then

$$xy \in I \Longrightarrow x \in I \text{ or } y \in I \Longrightarrow x + I = 0 \text{ or } y + I = 0.$$

Thus there are no zero divisors.
($\Leftarrow$) Likewise. □

Corollary

If I is maximal, then I is prime.

Proof

Note that every field R/I is an integral domain. □

Note. The converse is not true! For example,

$$R = \mathbb{F}[x, y], \qquad I = (y), \qquad R/I = \mathbb{F}[x].$$

Chinese Remainder Theorem

Let $I, J \subset R$ be ideals with

$$I + J = R$$

(so I and J are pairwise comaximal). Then:

1. The canonical homomorphism

$$\varphi : R \longrightarrow R/I \times R/J, \qquad a \longmapsto (a + I, a + J)$$

is surjective.
Side note. It is trivial to verify well-definedness and the homomorphism properties.

2.

$$R/(I \cap J) \cong R/I \times R/J.$$

Note. Coprimality in $\mathbb{N}$ gets generalized into the concept of comaximality. You should see why in the following proof.

Proof

For (1), since I, J are pairwise comaximal in R, there exist

$$a \in I, \qquad b \in J, \qquad a + b = 1 \in R.$$

Let

$$(\alpha + I, \beta + J) \in R/I \times R/J.$$

Consider $\alpha b + \beta a \in R$. We have

$$\alpha b + \beta a + I = \alpha(1 - a) + I = \alpha + I,$$
$$\alpha b + \beta a + J = \beta(1 - b) + J = \beta + J.$$

Therefore

$$\varphi(\alpha b + \beta a) = (\alpha + I, \beta + J),$$

so φ is surjective.
For (2), we claim that, for the φ in (1),

$$\ker \varphi = I \cap J.$$

The inclusion $\ker \varphi \supset I \cap J$ is obvious. Conversely, let $x \in \ker \varphi$. Then

$$x + I = I, \qquad x + J = J,$$

so $x \in I \cap J$. The isomorphism follows because φ is surjective and $I \cap J = \ker \varphi$, by applying the First Isomorphism Theorem. □

Generalized Chinese Remainder Theorem

Let $I_1, \ldots, I_n$ be pairwise comaximal ideals of a ring R; that is,

$$I_i + I_j = R \qquad \text{for every } i \neq j.$$

Then

$$R/\left(\bigcap_{i=1}^{n} I_i\right) \cong \prod_{i=1}^{n} R/I_i.$$

Proof by Induction

The base case $n = 2$ is the Chinese Remainder Theorem. For the induction step, suppose the isomorphism holds for $n - 1 \in \mathbb{N}$. We shall show it for $n \in \mathbb{N}$. Suppose $I_1, \ldots, I_n$ are mutually

Proof by Induction (continued)

comaximal. By the induction hypothesis,

$$R/(I_1 \cap \cdots \cap I_{n-1}) \cong R/I_1 \times \cdots \times R/I_{n-1}.$$

We claim that

$$(I_1 \cap I_2 \cap \cdots \cap I_{n-1}) + I_n = R.$$

Since

$$I_i + I_n = R \qquad (1 \leq i \leq n - 1),$$

there exist $a_i \in I_i$ and $b_i \in I_n$ such that

$$a_i + b_i = 1_R.$$

Multiplying gives

$$(a_1 + b_1)(a_2 + b_2) \cdots (a_{n-1} + b_{n-1}) = 1_R.$$

Expanding the product, we have

$$\underbrace{a_1 a_2 \cdots a_{n-1}}_{\in I_1 \cap \cdots \cap I_{n-1}} + \underbrace{\cdots}_{\substack{\text{terms containing some } b_i \in I_n, \\ \text{thus in } I_n \text{ since } I_n \text{ is an ideal}}} = 1_R \times 1_R \times \cdots \times 1_R = 1_R.$$

Thus

$$1_R \in I_1 \cap \cdots \cap I_{n-1} + I_n,$$

and hence

$$I_1 \cap \cdots \cap I_{n-1} + I_n = R.$$

Therefore, by the Chinese Remainder Theorem,

$$\begin{aligned} R/\big[(I_1 \cap \cdots \cap I_{n-1}) \cap I_n\big] &\cong R/(I_1 \cap \cdots \cap I_{n-1}) \times R/I_n \\ &\cong R/I_1 \times \cdots \times R/I_{n-1} \times R/I_n, \end{aligned}$$

where the second isomorphism is by the induction hypothesis. □

Notation. We may refer to the Cartesian product between two rings as a direct sum:

$$R_1 \times R_2 =: R_1 \oplus R_2.$$

The reason is that the two notions have different generalizations.

Example Let

$$R = \mathbb{Z}, \qquad I = (a), \qquad J = (b).$$

Then

$$I + J = (d), \qquad d = \gcd(a, b),$$

and

$$I \cap J = (m), \qquad m = \operatorname{lcm}(a, b).$$

If

$$I + J = R = \mathbb{Z},$$

then $d = 1$, hence $m = ab$. Therefore, by the Chinese Remainder Theorem,

$$R/(I \cap J) = R/(ab) = \mathbb{Z}/ab\mathbb{Z} \cong \mathbb{Z}/a\mathbb{Z} \oplus \mathbb{Z}/b\mathbb{Z}.$$

Thus, if

$$\begin{cases} x \equiv u \pmod{a}, \\ x \equiv v \pmod{b}, \end{cases}$$

then the system has a solution, and that solution is unique modulo ab, when $\gcd(a, b) = 1$.

Example Let $R = \mathbb{F}[x]$, where $\mathbb{F}$ is a field, and let

$$I = (f(x)), \qquad J = (g(x)), \qquad I + J = (\gcd(f, g)).$$

If $\gcd(f, g) = 1$, then

$$\mathbb{F}[x]/(f(x)g(x)) \cong \mathbb{F}[x]/(f(x)) \oplus \mathbb{F}[x]/(g(x)).$$

Groups

Definition

A *group* is a set G equipped with a law of composition, usually denoted multiplicatively,

$$\times : G \times G \longrightarrow G,$$

which satisfies:

1. there exists a neutral element;
2. for every $a \in G$, there exists $a^{-1} \in G$;
3. $\times$ is associative.

Definition

An *automorphism* of X is a map

$$\varphi : X \longrightarrow X$$

which preserves the structure on X.

Example

$$\begin{aligned} X \text{ is a set} &\implies \varphi \text{ is a bijection,} \\ X \text{ is a ring} &\implies \varphi \text{ is a ring isomorphism,} \\ X \text{ is a topological space} &\implies \varphi \text{ is a homeomorphism,} \\ X \text{ is a vector space} &\implies \varphi \text{ is a vector-space isomorphism.} \end{aligned}$$

Proposition

$\mathrm{Aut}(X)$ forms a group, where

$$\mathrm{Aut}(X) = \{\text{automorphisms of } X\}.$$

Formal Consequences

1. The identity is unique:
$$\widetilde{1} = \widetilde{1}\,1 = 1.$$
2. The inverse is unique. For every $g \in G$, if g' and g'' are inverses, then
$$g' = g'(gg'') = (g'g)g'' = g''.$$
3. *Cancellation law:* if $gx = gy$, where $g, x, y \in G$, then
$$x = y$$
by multiplying by g^{-1}.
4.
$$(g_1 g_2)^{-1} = g_2^{-1} g_1^{-1}.$$

Notation (additive notation). When the law of composition is written additively, the group is assumed to be abelian.

Definition

Let G be a group, $g \in G$, and $n \in \mathbb{N}$. Then

$$g^n := \underbrace{g \times g \times \cdots \times g}_{n},$$
$$g^{-n} := \underbrace{g^{-1} \times g^{-1} \times \cdots \times g^{-1}}_{n},$$
$$g^0 := 1.$$

Proposition

$$g^n \times g^m = g^{n+m}, \qquad (g^n)^m = g^{nm}.$$

Note. The only ring with $0 = 1$ is the zero ring. All other fields have

$$F^\times = F \setminus \{0\},$$

which must be an abelian group.

Notation.

$$R^\times := \{a \in R \mid a^{-1} \text{ exists in } R\}.$$

Examples

1. $G = \mathrm{Aut}(X)$.
2. $G = \{1\}$.
3. $(R, +, 0)$.
4. $(R^\times, \times, 1)$; for example,
$$\mathbb{Z}^\times = \{1, -1\}, \qquad \mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}.$$
5. If $\mathbb{F}$ is a field, then
$$M_n(\mathbb{F})^\times := GL_n(\mathbb{F}),$$
called the *general linear group*.
6. $$|(\mathbb{Z}/n\mathbb{Z})^\times| = \begin{cases} n-1, & n \text{ prime}, \\ \varphi(n), & \text{in general}, \end{cases}$$
where φ is Euler's totient function.

Finite Fields (Complementary Theorem)

Theorem

If R is a field and $|R| < \infty$, then

$$|R| = p^n$$

for some prime p and $n \geq 1$. Moreover, if

$$|\mathbb{F}_1| = |\mathbb{F}_2| < \infty,$$

then

$$\mathbb{F}_1 \cong \mathbb{F}_2.$$

Proof

We only prove the first claim; the second is a deeper result (Math 456/457, Algebra 3/Algebra 4). The first portion of the proof is the same. Suppose R is a field and $|R| < \infty$. There exists $1 \in R$. Consider the unique homomorphism

$$\varphi : \mathbb{Z} \longrightarrow R, \qquad 1 \longmapsto 1,$$

with the homomorphism defined recursively on $\mathbb{Z}$ by

$$\varphi(n) := \varphi(n-1) + \varphi(1).$$

We may verify that this is indeed a homomorphism. Then $\operatorname{Im}\varphi \subset R$ is a subring, and $\ker\varphi \subset \mathbb{Z}$ is an ideal. All ideals of $\mathbb{Z}$ are of the form

$$n\mathbb{Z} = (n).$$

Thus, by the First Isomorphism Theorem,

$$\mathbb{Z}/\ker\varphi = \mathbb{Z}/n\mathbb{Z} \cong \operatorname{Im}\varphi \subset R \qquad (n \in \mathbb{Z}).$$

Since $\operatorname{Im}\varphi \subset R$ is contained in a field, it has no zero divisors. Hence $\mathbb{Z}/n\mathbb{Z}$ has no zero divisors, so n is prime. Thus $\mathbb{Z}/n\mathbb{Z}$ is a field, and $\operatorname{Im}\varphi$ is a subfield of order n.
The ring R can be viewed as a vector space over the field $\operatorname{Im}\varphi$. It is finite-dimensional since $|R| < \infty$. Suppose

$$\dim R = m.$$

Since each vector in a vector space can be uniquely expressed as a linear combination of the basis with coefficients in the field (in this case $F = \operatorname{Im}\varphi$), the cardinality count continues on the next page.

Side result (Algebra 3). R must be generated by some $\alpha \in R$, and we have a surjective evaluation homomorphism

$$\mathrm{ev}_\alpha : (\mathbb{Z}/n\mathbb{Z})[x] \longrightarrow R.$$

By the First Isomorphism Theorem,

$$R \cong (\mathbb{Z}/n\mathbb{Z})[x]/\ker(\mathrm{ev}_\alpha),$$

where $\ker(\mathrm{ev}_\alpha) \subset (\mathbb{Z}/n\mathbb{Z})[x]$ is also an ideal.

Continuing the cardinality count,

$$\begin{aligned} |R| &= |\{a_1v_1 + \cdots + a_mv_m \mid a_i \in \operatorname{Im}\varphi\}| \\ &= |\operatorname{Im}\varphi| \cdot |\operatorname{Im}\varphi| \cdots |\operatorname{Im}\varphi| \\ &= |\operatorname{Im}\varphi|^m = p^m. \end{aligned}$$

Here $|\operatorname{Im}\varphi| = p$, since $\operatorname{Im}\varphi \cong \mathbb{Z}/n\mathbb{Z}$ and n is prime.

For the side result, recall that every ideal in $(\mathbb{Z}/n\mathbb{Z})[x]$ is principal. Therefore

$$\ker(\mathrm{ev}_\alpha) = (p(x))$$

for some $p(x) \in (\mathbb{Z}/n\mathbb{Z})[x]$. Suppose $\deg p(x) = m$. Then

$$(\mathbb{Z}/n\mathbb{Z})[x]/(p(x)) \cong R,$$

and hence

$$|(\mathbb{Z}/n\mathbb{Z})[x]/(p(x))| \cong |R|.$$

Thus

$$n^m = p^m = |R|, \qquad n = |\mathbb{Z}/n\mathbb{Z}| = p \text{ is prime.}$$

□

More Examples on Groups

1. **Cyclic groups.** If $g \in G$, then

$$\langle g \rangle = \{\ldots, g^{-3}, g^{-2}, g^{-1}, g^0 = 1, g, g^2, g^3, \ldots\}.$$

Case 1. All elements are distinct; that is, there is no $n \geq 1$ such that $g^n = 1$. Then there is a bijection

$$\mathbb{Z} \xrightarrow{\sim} \langle g \rangle, \qquad m \longmapsto g^m,$$

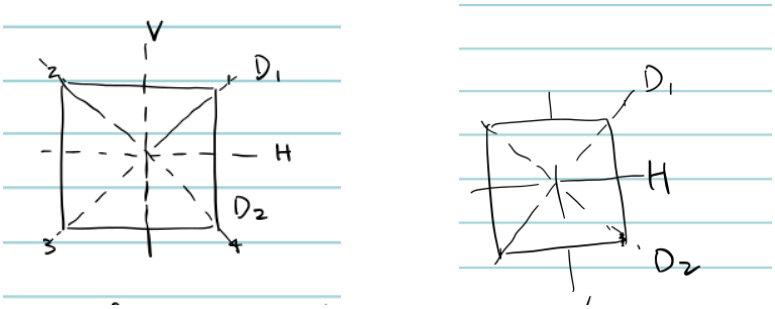
which is a group isomorphism:

$$\varphi : (\mathbb{Z}, +, 0) \longrightarrow \langle g \rangle, \qquad \varphi(n+m) = \varphi(n)\varphi(m).$$

Case 2. There exists a least $N \geq 1$ such that $g^N = 1$. Then there is a group isomorphism

$$\varphi : \mathbb{Z}/N\mathbb{Z} \xrightarrow{\sim} \langle g \rangle, \qquad [n] \longmapsto g^n.$$

2. **Dihedral group of order 8.** Let D_8 be the symmetries of a square. Let r be counterclockwise rotation by $\pi/2$ and let V, H, D_1, D_2 be the vertical, horizontal, and two diagonal reflections.



The vertices and the reflection axes V, H, D_1, D_2 of the square.

$$\langle r \rangle = \{1, r, r^2, r^3\},$$

and

$$D_8 = \{\underbrace{1, r, r^2, r^3}_{\text{rotations}}, \underbrace{V, H, D_1, D_2}_{\text{reflections}}\}.$$

Its multiplication table is

$\cdot$	1	r	r^2	r^3	V	H	D_1	D_2
1	1	r	r^2	r^3	V	H	D_1	D_2
r	r	r^2	r^3	1	D_2	D_1	V	H
r^2	r^2	r^3	1	r	H	V	D_2	D_1
r^3	r^3	1	r	r^2	D_1	D_2	H	V
V	V	D_1	H	D_2	1	r^2	r	r^3
H	H	D_2	V	D_1	r^2	1	r^3	r
D_1	D_1	H	D_2	V	r^3	r	1	r^2
D_2	D_2	V	D_1	H	r	r^3	r^2	1

Composition convention. Products of symmetries are evaluated from right to left.

3. **Dihedral group of order $2n$.**

$$D_{2n} = \{\underbrace{1, r, \ldots, r^{n-1}}_{r=\text{rotation counterclockwise by } 2\pi/n}, \underbrace{D_1, \ldots, D_n}_{\text{reflections}}\}.$$

4. **Symmetric group.**

$$X = \{1, 2, \ldots, n\}, \qquad S_n = \mathrm{Aut}(X) = \{f : X \to X \mid f \text{ is bijective}\}.$$

For $n = 3$,

$$S_n = S_3 = \{\underbrace{\mathrm{id}}_{1}, \underbrace{(23), (12), (13)}_{\text{transpositions}}, \underbrace{(123), (132)}_{\text{3-cycles}}\}.$$

In general,

$$|S_n| = n! = {}_n\mathrm{Perm}_n\,.$$

The law of composition is composition of bijective maps. For example,

$$(23) \circ (13) = (123).$$

The maps act as

$$(23) : \begin{cases} 2 \mapsto 3, \\ 3 \mapsto 2, \\ 1 \mapsto 1, \end{cases} \qquad (13) : \begin{cases} 1 \mapsto 3, \\ 3 \mapsto 1, \\ 2 \mapsto 2, \end{cases}$$

so their composition acts as

$$1 \mapsto 2, \qquad 3 \mapsto 1, \qquad 2 \mapsto 3.$$

Cycle Notation

Example In S_4,

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix} = (1324),$$

a 4-cycle. Immediately, the element has order 4.

Example

$$(1324)^2 = (12)(34),$$

a product of two 2-cycles (transpositions), and

$$(1324) \circ (14)(23) = (1)(2)(34) = (34).$$

Note. This gives a neat way to find the order of $P \in S_n$:

$$|\langle P \rangle| = \text{lcm}(\text{distinct cycle lengths}).$$

For example,

$$|\langle (123)(45) \rangle| = \text{lcm}(2,3) = 6.$$

Revisit $GL_n(\mathbb{F})$

Question. What is

$$|GL_n(\mathbb{Z}/p\mathbb{Z})|?$$

Answer. Construct an invertible matrix row by row.

The first row cannot be zero, so there are

$$p^n - 1$$

choices.

The second row cannot be a linear combination of the first row, so there are

$$p^n - p$$

choices.

The third row cannot be a linear combination of the first and second rows, which has the form $ar_1 + br_2$. There are $p \times p$ different possibilities for the coefficients a and b, so there are

$$p^n - p^2$$

choices.

Continuing in this way, the nth row has

$$p^n - p^{n-1}$$

possibilities, in order to avoid being a linear combination of the previous $n-1$ rows. Thus, in total,

$$|GL_n(\mathbb{Z}/p\mathbb{Z})| = \prod_{i=0}^{n-1} (p^n - p^i) = p^{\frac{n(n-1)}{2}} (p^n - 1)(p^{n-1} - 1) \cdots (p-1).$$

Conjugacy

Definition

Let G be a group. Elements $g_1, g_2 \in G$ are *conjugate* if there exists $h \in G$ such that

$$g_2 = h g_1 h^{-1}.$$

We write $g_1 \sim g_2$.

Note. If G is abelian, then conjugacy is equality.

Proposition

Conjugacy is an equivalence relation.

Definition

Conjugacy classes are the equivalence classes arising from this relation.

Quaternions.

$$Q = \{\pm 1, \pm i, \pm j, \pm k\},$$

where

$$i = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad 1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad j = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

They satisfy

$$i^2 = j^2 = k^2 = -1, \qquad ij = -ji = k.$$

Claim.

Conjugacy preserves order:

$$(hgh^{-1})^n = (hgh^{-1}) \cdots (hgh^{-1}) = hg^n h^{-1}.$$

If hgh^{-1} has order n, then g must have order n.

Remark This gives a quick way of identifying conjugacy classes, for example in S_3. We can immediately infer that

$$(123) \not\sim (12),$$

since they have distinct orders. Moreover,

$$S_3 = \{\underbrace{1}, \underbrace{(12), (23), (31)}, \underbrace{(123), (132)}\},$$

giving three conjugacy classes in S_3.

Theorem

If $g_1, g_2 \in S_n$, then

$$g_1 \sim g_2 \qquad \Longleftrightarrow \qquad g_1, g_2 \text{ have the same cycle structure.}$$

Example The permutations

$$(123)(45) \sim (13)(254)$$

both have cycle shape $2, 3$ (a partition of 5, since $2 + 3 = 5$), whereas

$$(123)(45) \not\sim (13)(24),$$

because the latter has cycle shape $2, 2, 1$, another partition of 5.

Proof

($\Rightarrow$) Suppose

$$g_2 = hg_1h^{-1}.$$

Suppose $g_1(i) = j$, where $i, j \in \{1, \ldots, n\}$. Then

$$g_2(h(i)) = hg_1h^{-1}h(i) = hg_1(i) = h(j).$$

If

$$g_1 = (i_1 i_2 \cdots i_k)(j_1 \cdots j_r) \cdots,$$

then, from the equality above,

$$g_2 = (h(i_1)h(i_2) \cdots h(i_k))(h(j_1) \cdots h(j_r)) \cdots.$$

In words, however g_1 acts upon $\{1, \ldots, n\}$, g_2 does exactly the same to

$$\{h(1), h(2), \ldots, h(n)\}.$$

($\Leftarrow$) Suppose $g_1, g_2 \in S_n$ have the same cycle shape, with

$$g_1 = (i_1 i_2 \cdots i_k) \cdots, \qquad g_2 = (j_1 j_2 \cdots j_k) \cdots.$$

Define

$$h : \{1, \ldots, n\} \longrightarrow \{1, \ldots, n\}$$

by

$$h(i_1) = j_1, \qquad h(i_2) = j_2, \qquad \ldots, \qquad h(i_k) = j_k, \qquad \ldots.$$

Thus

$$g_2 = (h(i_1)h(i_2) \cdots h(i_k)) \cdots.$$

Then

$$hg_1 = g_2h \qquad \Longrightarrow \qquad g_1 = h^{-1}g_2h \qquad \Longrightarrow \qquad g_1 \sim g_2.$$

□

Question. We saw that order is preserved by conjugation. Is the converse true in S_n; that is, does

$$\text{same order} \Longrightarrow \text{conjugate?}$$

Answer. Yes in S_3. No if $n \geq 4$. In S_4,

$$(12) \not\sim (12)(34),$$

because their cycle structures are different, even though their orders are the same.

Example Let

$$G = GL_n(\mathbb{F}), \qquad M_1, M_2 \in G.$$

Then $M_1 \sim M_2$ means that there exists $P \in GL_n(\mathbb{F})$ such that

$$M_2 = PM_1P^{-1}.$$

Invariants under conjugacy (similar matrices) include determinant, trace, characteristic polynomial, minimal polynomial, and Jordan normal form.

Homomorphisms and Isomorphisms

Definition

A map

$$\varphi : G_1 \longrightarrow G_2$$

is a homomorphism if

$$\varphi(gh) = \varphi(g)\varphi(h).$$

Consequences.

$$\varphi(1_{G_1}) = 1_{G_2}, \qquad \varphi(g^{-1}) = [\varphi(g)]^{-1}.$$

Definition

$$\ker \varphi = \{g \in G_1 \mid \varphi(g) = 1_{G_2}\}.$$

Formal consequence. The kernel of any ring homomorphism can be viewed as the kernel of the additive group of the ring.

Subgroups and Lagrange's Theorem

Theorem

Let G be a finite group and H a subgroup of G. Then

$$|H| \mid |G|.$$

Definition

Define

$$g_1 \sim_H g_2 \qquad \Longleftrightarrow \qquad \exists h \in H \text{ such that } g_2 = g_1 h.$$

Proposition

$\sim_H$ is an equivalence relation.

Definition

An equivalence class for $\sim_H$, denoted $[g]_H$, is called a *(right) coset* of H in G. Notice that

$$[g]_H = \{gh \mid h \in H\}.$$

Thus we denote it by gH.
The quotient is

$$G/H = \{gH \mid g \in G\}.$$

Proof of Lagrange's Theorem

Note that

$$G = \coprod_{gH \in G/H} gH, \qquad |gH| = |H|.$$

Therefore

$$|G| = \sum_{gH \in G/H} |gH| = |G/H|\,|H|.$$

This is the counting formula. □

Example Let

$$G = (\mathbb{Z}, +, 0), \qquad H = 2\mathbb{Z} \leq G.$$

Then

$$G/H = \{2\mathbb{Z}, 2\mathbb{Z}+1\} = \mathbb{Z}/2\mathbb{Z},$$

where the displayed sets are the right cosets of H.

Example Let

$$G = D_8 = \{1, r, r^2, r^3, H, V, D_1, D_2\}.$$

Take

$$H_1 = \{1, r, r^2, r^3\} \leq G.$$

Then

$$VH_1 = \{V, Vr = D_1, Vr^2 = H, Vr^3 = D_2\},$$

and

$$D_8/H_1 = \{H_1, VH_1\} = \{\{\text{rotations}\}, \{\text{reflections}\}\}.$$

Also take

$$H_2 = \{1, V\} \leq G.$$

Its right cosets are

$$\begin{aligned} H_2 &= \{1, V\}, \\ rH_2 &= \{r, D_2\}, \\ r^2H_2 &= \{r^2, H\}, \\ r^3H_2 &= \{r^3, D_1\}. \end{aligned}$$

Example Let

$$G = S_n, \qquad H = S_{n-1} \leq S_n.$$

Then

$$|G/H| = \frac{n!}{(n-1)!} = n.$$

Moreover,

$$G/H = \{gH \mid g \in G\}.$$

Consider

$$\varphi : G/H \longrightarrow \{1, \ldots, n\} \qquad \text{such that}$$

Consider the map

$$G/H \longrightarrow \{1, \ldots, n\}, \qquad gH \longmapsto g'(n),$$

where $g' = gh$ for any $h \in H$.

The map φ is well-defined since every permutation $h \in H = S_{n-1} \leq G$ fixes the n-th element and permutes the first $n-1$ elements. Moreover, φ is bijective. Surjectivity is trivial. For injectivity, suppose $\varphi(g_1H) = \varphi(g_2H)$. We need to show that $g_1H = g_2H$, that is, that there is an $h \in H$ such that $g_1 = g_2h$. We have

$$g_1h_1(n) = g_2h_2(n) \implies g_1(n) = g_2(n).$$

Choose $h = g_1^{-1}g_2$. Then

$$h(n) = g_1^{-1}g_2(n) = n,$$

so $h \in H$, and $h = g_1^{-1}g_2$ implies $g_1h = g_2$.

$$\boxed{g_1 : n \mapsto g_1(n), \qquad n \xleftarrow{g_2^{-1}} g_2(n)}$$

Remark There is no natural group law

$$G/H \times G/H \longrightarrow G/H$$

unless H is normal.

Example Let $G = D_8$ and $H_3 = \{1, V\} \leq G$. The tempting multiplication is not well-defined (there are four different products in G/H_3).

However, there is a natural multiplication

$$G \times G/H \longrightarrow G/H, \qquad (g, aH) \longmapsto (ga)H.$$

Group Actions

Definition

An action of a group G on a set X is a map

$$\cdot : G \times X \longrightarrow X, \qquad (g, x) \longmapsto gx,$$

such that, for all $g_1, g_2 \in G$ and $x \in X$,

$$1_Gx = x, \qquad g_1(g_2x) = (g_1g_2)x.$$

In this case we say that X is a G-set.

Example (Ways of thinking about group actions.) Suppose $G \times X \to X$ is a group action.

1. We obtain a homomorphism

$$\varphi : G \longrightarrow \mathrm{Aut}(X) = \mathrm{Perm}(X) = S_X \cong S_n \quad \text{if } |X| = n,$$

given by

$$\varphi(g) : X \longrightarrow X, \qquad x \longmapsto gx.$$

We have several things to check. First, $\varphi(g) \in \mathrm{Aut}(X)$ for every $g \in G$. For $x \in X$,

$$\varphi(g)(g^{-1}x) = g(g^{-1}x) = (gg^{-1})x = 1x = x,$$

so $\varphi(g)$ is surjective.

For injectivity, let $x_1, x_2 \in X$ satisfy $\varphi(g)(x_1) = \varphi(g)(x_2)$, that is, $gx_1 = gx_2$. Then

$$g^{-1}(gx_1) = g^{-1}(gx_2) \implies x_1 = 1x_1 = (g^{-1}g)x_1 = (g^{-1}g)x_2 = 1x_2 = x_2.$$

Therefore $\varphi(g)$ is a bijection and belongs to $\mathrm{Aut}(X)$.

Second, φ is a group homomorphism. Let $g_1, g_2 \in G$. Then

$$\begin{array}{rcl} \varphi(g_1g_2) : X & \longrightarrow & X \\ x & \longmapsto & (g_1g_2)x \end{array} \qquad \begin{array}{rcl} \varphi(g_1)\varphi(g_2) : X & \longrightarrow & X \\ x & \longmapsto & g_1(g_2x) = (g_1g_2)x. \end{array}$$

Thus

$$\varphi(g_1g_2) = \varphi(g_1)\varphi(g_2), \qquad g_1, g_2 \in G.$$

Cayley's Theorem

Theorem

If G is a finite group, then G is isomorphic to a subgroup of S_n for some $n \in \mathbb{N}$.

Proof

If $H \leq G$, then there is an action

$$G \times G/H \longrightarrow G/H, \qquad (g, aH) \longmapsto (ga)H.$$

Hence there is a homomorphism

$$\varphi : G \longrightarrow S_{G/H}, \qquad g \longmapsto (\varphi(g) : G/H \longrightarrow G/H, \quad aH \longmapsto gaH)\,.$$

If φ is injective, then $\varphi(G) \leq S_{G/H}$. Thus the question is: when is $\varphi : G \to \mathrm{Aut}(G/H)$ injective?

$$\begin{aligned} g \in \ker\varphi &\iff g \in G \text{ and } \varphi(g) = 1_{\mathrm{Aut}(G/H)} \\ &\iff gaH = aH \quad \text{for every } aH \in G/H \\ &\iff gaH = aH \quad \text{for every } a \in G \\ &\iff ga \sim_H a \quad \text{for every } a \in G \\ &\iff ga = ah \text{ for some } h \in H \quad \text{for every } a \in G \\ &\iff a^{-1}ga \in H \quad \text{for every } a \in G \\ &\iff g \in \bigcap_{a \in G} aHa^{-1}. \end{aligned}$$

Therefore

$$\ker\varphi = \bigcap_{a \in G} aHa^{-1}.$$

Taking $H = \{1_G\}$ concludes the proof: then

$$\ker\varphi = \bigcap_{a \in G} a\{1\}a^{-1} = \{1\},$$

so φ is injective. Hence

$$G \cong \varphi(G) \leq S_{G/H} = S_{G/\{1\}} = S_G = S_{|G|}.$$

Remark This proof of Cayley's theorem works, but is not efficient in practice.

Example For $G = S_n$ and $H = \{1\}$, the construction gives

$$G \cong \varphi(G) \leq S_{G/H} \cong S_{S_n}, \qquad |S_{S_n}| = (n!)!.$$

Better. Let $G = S_n$ and $H = S_{n-1}$. Define

$$\varphi : G \longrightarrow \mathrm{Aut}(G/H), \qquad g \longmapsto (\varphi(g) : G/H \longrightarrow G/H, \quad aH \longmapsto gaH)\,.$$

Then

$$G/H \cong \{1, 2, \ldots, n\}, \qquad G \cong \varphi(G) \leq S_{G/H} \cong S_n.$$

The subgroup has the same size as S_n:

$$n! = |S_n| = |G| = |\varphi(G)| = |S_n| = n!,$$

so the subgroup must be the whole group.

Question. When is the natural law of composition

$$G/H \times G/H \longrightarrow G/H$$

well-defined?

Multiplication is well-defined if

$$g^{-1}Hg \leq H \qquad \text{for every } g \in G.$$

Definition

H is a *normal subgroup*, written $H \trianglelefteq G$, if

$$g^{-1}Hg \leq H.$$

In fact $g^{-1}Hg = H$ for every g, because $gHg^{-1} = H$, and $H \to g^{-1}Hg$ is a bijection.

Example For $G = S_3$,

$$\{1\}, \quad G, \quad \{1, (123), (132)\}$$

are normal in G, while

$$\{1, (13)\}, \quad \{1, (12)\}, \quad \{1, (23)\}$$

are not normal in G.

Example Let $G = D_8$ and recall the conjugacy classes

$$\{1\}, \quad \{r^2\}, \quad \{r, r^3\}, \quad \{V, H\}, \quad \{D_1, D_2\}.$$

The subgroups, arranged by order, are

$\lvert H\rvert$	H
1	$\underline{\{1\}}$
2	$\underline{\{1, V\}}$, $\{1, H\}$, $\{1, D_1\}$, $\{1, D_2\}$, $\underline{\{1, r^2\}}$
4	$\underline{\{1, r, r^2, r^3\}}$, $\underline{\{1, r^2, V, H\}}$, $\underline{\{1, r^2, D_1, D_2\}}$
8	$\underline{D_8}$.

The underlined entries are the normal subgroups.

Remark If $H \leq G$ and $|H| = |G|/2$, then H must be normal.

Kernel

Proposition

The kernel of a homomorphism is a normal subgroup of the domain group.

Question. Given $H \trianglelefteq G$, is H the kernel of some homomorphism $\varphi : G \to ?$

Answer. Take $? = G/H$, since $H \trianglelefteq G$ (First Isomorphism Theorem).

Ring	Group
abelian group plus multiplication	group
ideals: subgroups	normal subgroups: subgroups
stable under multiplication	stable under conjugation

First Isomorphism Theorem for Groups

Theorem

If $\varphi : G \to G'$ is a group homomorphism, then there is an isomorphism

$$\overline{\varphi} : G/\ker\varphi \xrightarrow{\sim} \operatorname{Im}\varphi \leq G'.$$

Proof

Define $\overline{\varphi}(gH) = \varphi(g)$. Prove that $\overline{\varphi}$ is well-defined, injective, surjective, and a homomorphism.

Problem: Classification of Finite Groups?

Proposition

Let G be a group. If $g \in G$ has finite order N, then

$$\langle g \rangle \cong \mathbb{Z}/N\mathbb{Z}$$

via $\varphi(g^n) = n$. If g has order 0, then

$$\langle g \rangle \cong \mathbb{Z}.$$

Here $\varphi(ab) = \varphi(a) + \varphi(b)$, since $\mathbb{Z}/N\mathbb{Z}$ and $\mathbb{Z}$ are additive groups.

Corollary

If G is cyclic, then

$$G \cong \mathbb{Z}/|G|\mathbb{Z}, \qquad \mathbb{Z}/\infty\mathbb{Z} = \mathbb{Z}.$$

Proposition

Every group of prime order is cyclic.

Proof

$|\langle g \rangle|$ divides $|G|$. Thus $\langle g \rangle = G$, since $|G|$ is prime.

Groups of Small Order

$\lvert G\rvert$	G
1	1
2	$\mathbb{Z}/2\mathbb{Z}$
3	$\mathbb{Z}/3\mathbb{Z}$
4	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ or $\mathbb{Z}/4\mathbb{Z}$
5	$\mathbb{Z}/5\mathbb{Z}$
6	$\mathbb{Z}/6\mathbb{Z}$ or S_3.

Klein Four Group. It has the presentation

$$\{1, \sigma, \tau, \sigma\tau\}, \qquad \sigma^2 = \tau^2 = 1, \quad \sigma\tau = \tau\sigma,$$

and the realizations

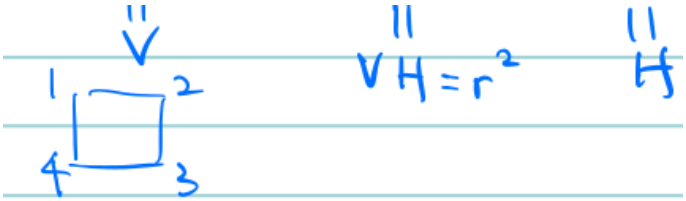
$$\left\{ \begin{bmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{bmatrix} \right\},$$

$$\{1, (12), (34), (12)(34)\},$$

and

$$\{1, (12)(34), (13)(24), (14)(23)\}.$$

The last realization is the subgroup of S_4 obtained from the symmetries of a square. Its three nonidentity elements are the vertical reflection V, the horizontal reflection H, and the half-turn r^2.



The square realization of the Klein four group.

Recall Cayley's theorem: every finite group can be isomorphically embedded as a subgroup of S_n for some $n \in \mathbb{N}$.

Proposition

$$V = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \ncong \mathbb{Z}/4\mathbb{Z} = C_4.$$

Proof

If two groups are isomorphic, then the elements of each particular order must be the same in number.

Proposition

If $|G| = 4$, then G is cyclic (and hence isomorphic to C_4) or isomorphic to

$$V \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \cong C_2 \times C_2.$$

Proof

Either there is an element of order 4, or all nonidentity elements are of order 2. By Lagrange's theorem, the latter group is isomorphic to V (Klein four).

Claim.

$$S_3 \ncong \mathbb{Z}/6\mathbb{Z} \cong C_6.$$

Proof

Count elements of order 6.

Remark

$$\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}.$$

Since $\gcd(2,3) = 1$, the Chinese remainder theorem gives the natural ring isomorphism above, which also yields an isomorphism of the underlying additive groups.

Question. Are there other groups of order 6?

Recall. A group action of G on X is a map $G \times X \to X$ such that

$$(1_G, x) \longmapsto x, \qquad (g_1, (g_2, x)) \longmapsto (g_1 g_2, x).$$

We write $(g, x) \mapsto gx \in X$. The set X is called a G-set, or a permutation representation of G.

Remark The notion of an action is equivalent to a homomorphism from G to $\mathrm{Aut}(X)$, the set of permutations of X:

$$\varphi : G \longrightarrow \mathrm{Aut}(X), \qquad g \longmapsto (\varphi(g) : X \to X,\ x \mapsto gx)\,.$$

The homomorphism property follows from the group-action axioms.

Example If $G = S_n$ and $X = \{1, \ldots, n\}$, then

$$\varphi : G \longrightarrow \mathrm{Aut}(X) = S_n, \qquad \varphi = \mathrm{id}\,.$$

Example If $X = G$, we get a homomorphism

$$i : G \longrightarrow S_G.$$

We may verify $\ker(i) = \{1\}$, so

$$G \cong \mathrm{Im}(i) \leq S_G,$$

which is Cayley's theorem.

Example If H is a subgroup of G, then

$$X = G/H = \{gH : g \in G\}$$

is a G-set under

$$G \times G/H \longrightarrow G/H, \qquad (g, aH) \longmapsto gaH.$$

Definition

Let G be a group and let X_1, X_2 be G-sets. We say $X_1 \cong X_2$ as G-sets if there is an isomorphism f such that

$$f(gx) = gf(x) \qquad \text{for every } x \in X_1, g \in G.$$

Definition

A G-set X is *transitive* if, for every $x, y \in X$, there is a $g \in G$ such that $gx = y$.

Theorem

Every transitive G-set is isomorphic, as a G-set, to G/H for some subgroup $H \leq G$.

Proof

Choose $x_0 \in X$ and set

$$H = \mathrm{Stab}(x_0) := \{g \in G : gx_0 = x_0\},$$

the stabilizer of x_0.

Proof (continued)

Define

$$\eta : G/H \longrightarrow X, \qquad aH \longmapsto ax_0.$$

We claim that η is well-defined and is a G-set isomorphism. If $a_1H = a_2H$, then $a_1 = a_2h$ for some $h \in H$, and hence

$$a_1x_0 = a_2hx_0 = a_2x_0,$$

because $h \in \mathrm{Stab}(x_0) = H$.
For injectivity, suppose $g_1, g_2 \in G$ and $\eta(g_1H) = \eta(g_2H)$. Then

$$\begin{aligned} g_1x_0 &= g_2x_0, \\ g_1^{-1}(g_1x_0) &= g_1^{-1}(g_2x_0), \\ (g_1^{-1}g_1)x_0 &= (g_1^{-1}g_2)x_0, \\ x_0 &= (g_1^{-1}g_2)x_0, \end{aligned}$$

so $g_1^{-1}g_2 \in H$, equivalently $g_1H = g_2H$.
For surjectivity, given $y \in X$, transitivity gives a $g \in G$ such that $gx_0 = y$. Then $\eta(gH) = gx_0 = y$.
Finally, η preserves the G-set structure. Given $g \in G$ and $aH \in G/H$,

$$\eta\big(g(aH)\big) = \eta(gaH) = gax_0 = g\eta(aH).$$

Remark This allows us to find all possible transitive G-sets on which G can act.

Example (Stabilizer; transitive G-set.) Let

$$G = S_n, \qquad X = \{1, 2, \ldots, n\}, \qquad x_0 = n.$$

Then $\mathrm{Stab}(n) = S_{n-1}$, since $gx_0 = x_0$, and the preceding theorem gives

$$G/H = S_n/S_{n-1} \cong X = \{1, \ldots, n\}.$$

Example (Transitive G-set.) Check whether

$$(12345)S_4 = (451)S_4 \qquad \text{in } S_5.$$

Under the preceding isomorphism,

$$\eta\big((12345)S_4\big) = (12345)5 = 1,$$

and

$$\eta\big((451)S_4\big) = (451)5 = 1.$$

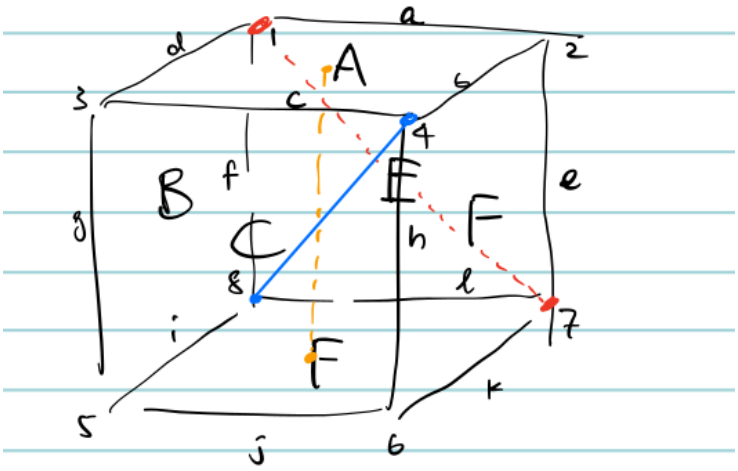
Since η is an isomorphism between S_5/S_4 and $X = \{1, 2, \ldots, 5\}$, equal images imply

$$(12345)S_4 = (451)S_4.$$

Example Let X be a cube and let G be its group of orientation-preserving rotations. Set

$$X_0 = \{\text{the 8 vertices}\}, \qquad X_1 = \{\text{the 12 edges}\}, \qquad X_2 = \{\text{the 6 faces}\}.$$

These are transitive G-sets.



Labels for the vertices, edges, faces, and the three representative rotation axes.

With the labels in the diagram,

$$\begin{aligned} H_0 &= \mathrm{Stab}(8) \\ &= \left\{ \begin{array}{c} \text{rotations about the axis } \overline{48} \\ \text{by multiples of } \dfrac{2\pi}{3} \end{array} \right\} \\ &\cong \mathbb{Z}/3\mathbb{Z}\ (C_3), \end{aligned}$$

$$H_1 = \mathrm{Stab}(\ell) = \{\text{rotations about the axis } \overline{17} \text{ by multiples of } \pi\} \cong \mathbb{Z}/2\mathbb{Z}\ (C_2),$$

and

$$\begin{aligned} H_2 &= \mathrm{Stab}(A) \\ &= \left\{ \begin{array}{c} \text{rotations about the axis through the centers of faces } A \text{ and } F \\ \text{by multiples of } \dfrac{\pi}{2} \end{array} \right\} \\ &\cong \mathbb{Z}/4\mathbb{Z}\ (C_4). \end{aligned}$$

Thus, by the orbit–stabilizer counting formula,

$$|G| = |X_0|\,|\,\mathrm{Stab}(8)| = |X_1|\,|\,\mathrm{Stab}(\ell)| = |X_2|\,|\,\mathrm{Stab}(A)| = 24.$$

Orbit–Stabilizer Theorem

If X is a transitive G-set and $H = \mathrm{Stab}(x_0)$ for $x_0 \in X$, then

$$|G| = |H|\,|X|.$$

The proof follows from $G/H \cong X$ and the counting formula.

Definition

Let X be a set on which G acts. For $x \in X$, the orbit of x is

$$\mathrm{Orb}(x) := Gx \subset X.$$

Proposition

The orbits of an action of G on X form a partition of X, or equivalently an equivalence relation on X.

Proof

Define

$$x \sim_{\mathrm{orb}} y \iff x \in \mathrm{Orb}(y).$$

We prove that this is an equivalence relation, so that it partitions X.

Reflexivity. We have $x \sim_{\mathrm{orb}} x$ by the group-action axiom $1x = x$.

Symmetry. If $x \sim_{\mathrm{orb}} y$, then $x \in \mathrm{Orb}(y)$, so $gy = x$ for some $g \in G$. Hence

$$g^{-1}(gy) = g^{-1}x \implies (g^{-1}g)y = g^{-1}x \implies y \in \mathrm{Orb}(x),$$

and therefore $y \sim_{\mathrm{orb}} x$.

Transitivity. If $x \sim_{\mathrm{orb}} y$ and $y \sim_{\mathrm{orb}} z$, then $x = g_1 y$ and $y = g_2 z$ for some $g_1, g_2 \in G$. Thus

$$x = g_1 g_2 z \in \mathrm{Orb}(z),$$

so $x \sim_{\mathrm{orb}} z$.

Remark We can restate the orbit–stabilizer theorem to justify its name:

$$|G| = |H|\,|X| = |\,\mathrm{Stab}_G(x_0)|\,|\,\mathrm{Orb}(x_0)|,$$

for every $x_0 \in X$, when X is a transitive G-set.

Proposition

If a group G acts on X, then every equivalence class for $\sim_{\mathrm{orb}}$ is itself a set on which G acts transitively.

Proof

Every equivalence class is of the form $\mathrm{Orb}(x)$. Since $\sim_{\mathrm{orb}}$ is an equivalence relation,

$$G \times \mathrm{Orb}(x) \longrightarrow \mathrm{Orb}(x), \qquad (g, y) \longmapsto gy,$$

is closed under the group action. Thus $\mathrm{Orb}(x)$ is a G-set under the restriction of the action on X, and transitivity is inherited by each orbit.

Question. Given a G-set X, write it as a disjoint union of transitive G-sets (the orbits of the action):

$$X = \bigsqcup_{i=1}^{t} X_i.$$

What is the number t?

Cauchy–Frobenius Theorem

Also called the Cauchy–Frobenius lemma or Burnside's lemma. Define

$$\mathrm{FP}_X : G \longrightarrow \mathbb{N}, \qquad g \longmapsto \#\{\text{fixed points of the action of } g \text{ on } X\}.$$

Then

$$\boxed{\frac{1}{|G|}\sum_{g\in G} \mathrm{FP}_X(g) = t = \#\{\text{disjoint orbits / transitive } G\text{-sets}\}.}$$

Proof

The proof follows mechanically from the orbit–stabilizer theorem:

$$|G| = |\operatorname{Orb}(x)|\,|\operatorname{Stab}(x)|.$$

Note that

$$\begin{aligned}\sum_{g\in G} \mathrm{FP}_X(g) &= \sum_{g\in G} \#\{x \in X : gx = x\} \\ &= \#\{(g,x) \in G \times X : gx = x\} \\ &= \sum_{x\in X} \#\{g \in G : gx = x\} \\ &= \sum_{x\in X} |\operatorname{Stab}_G(x)|.\end{aligned}$$

Let $x_1, \ldots, x_t$ be representatives of the t disjoint orbits. Since $x \in \operatorname{Orb}(y)$ implies $\operatorname{Orb}(x) = \operatorname{Orb}(y)$, the orbit–stabilizer theorem gives $|\operatorname{Stab}(x)| = |\operatorname{Stab}(y)|$. Therefore

$$\begin{aligned}|\operatorname{Orb}(x)|\,|\operatorname{Stab}(x)| \cdot t &= |\operatorname{Orb}(x_1)|\,|\operatorname{Stab}(x_1)| + \cdots + |\operatorname{Orb}(x_t)|\,|\operatorname{Stab}(x_t)| \\ &= \sum_{x\in\operatorname{Orb}(x_1)} |\operatorname{Stab}(x_1)| + \cdots + \sum_{x\in\operatorname{Orb}(x_t)} |\operatorname{Stab}(x_t)| \\ &= \sum_{x\in X} |\operatorname{Stab}(x)| = \sum_{g\in G} \mathrm{FP}_X(g).\end{aligned}$$

Remark 1. The Cauchy–Frobenius theorem and the orbit–stabilizer theorem both generalize to arbitrary G-sets X, not necessarily transitive. The orbit–stabilizer theorem applies because each orbit of the action is a transitive G-set.

Remark 2. If $x \sim_{\text{orb}} y$, then

$$\operatorname{Stab}_G(x) \cong_{\text{conj}} \operatorname{Stab}_G(y),$$

and hence their cardinalities are equal.

Example Let $G = S_4$ and $X = \{1, 2, 3, 4\}$. How many orbits are there?

Note.

$$\mathrm{FP}_X(g) = \mathrm{FP}_X(hgh^{-1}) \qquad \text{for every } h \in G.$$

Conjugate elements have the same cycle structure and act on X in the same way, up to a change of indices.

Conjugacy Classes in S_4

	cycle structure	conjugacy class	cardinality
C_1	4	$(1234), (1243), (1324), (1342), (1423), (1432)$	6
C_2	$3+1$	$(123), (132), (124), (142), (134), (143), (234), (243)$	8
C_3	$2+1+1$	$(12), (23), (24), (13), (34), (14)$	6
C_4	$2+2$	$(12)(34), (13)(24), (14)(23)$	3
C_5	$1+1+1+1$	1	1

The class equation is

$$|S_4| = 1 + 3 + 6 + 6 + 8.$$

For $X = \{1, 2, 3, 4\}$, it suffices to use one representative from each class:

C	$\#C$	$\mathrm{FP}_X(g)$
1	1	4
$(12)(34)$	3	0
(12)	6	2
(123)	8	1
(1234)	6	0.

Thus

$$\#\mathcal{O} = \frac{1}{|G|} \sum_{g \in G} \mathrm{FP}_X(g) = \frac{1}{24}(1 \cdot 4 + 0 + 6 \cdot 2 + 8 \cdot 1 + 0) = 1.$$

For $X \times X$, where G acts componentwise on ordered pairs,

$$\mathrm{FP}_{X\times X}(g) = \mathrm{FP}_X(g)\,\mathrm{FP}_X(g) = \big(\mathrm{FP}_X(g)\big)^2,$$

because $g(x_1, x_2) = (gx_1, gx_2) = (x_1, x_2)$. Hence

$$\#\mathcal{O} = \frac{1}{24}\left(1 \cdot 4^2 + 0 + 6 \cdot 2^2 + 8 \cdot 1^2 + 0\right) = 2.$$

The orbit decomposition is

$$X \times X = \{(1,1), (2,2), (3,3), (4,4)\} \sqcup \{(i,j) : i \neq j\},$$

the two transitive G-sets.

For X^3,

C	$\#C$	$\mathrm{FP}_X(g)$	$\mathrm{FP}_{X^3}(g)$
1	1	4	$4^3 = 64$
$(12)(34)$	3	0	0
(12)	6	2	$2^3 = 8$
(123)	8	1	$1^3 = 1$
(1234)	6	0	0.

Therefore

$$\#\mathcal{O} = \frac{1}{24}(64 + 48 + 8) = 5.$$

The orbit decomposition has the five patterns

$$X^3 = \{(a,a,a)\} \sqcup \{(a,a,b)\} \sqcup \{(a,b,a)\} \sqcup \{(b,a,a)\} \sqcup \{(a,b,c)\}.$$

Aut(**cube**)

There are three axes through opposite face centers (type I), six axes through opposite edge centers (type II), and four axes through opposite vertices (type III).



The three types of rotational axes of a cube.

The rotations of the cube can be counted as follows:

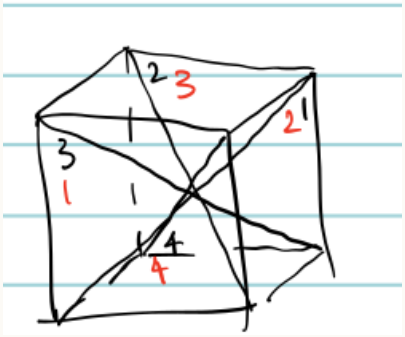
elements of $\mathrm{Aut}(C)$	order	#
id	1	1
r_f	4	6
r_f^2	2	3
r_e	2	6
r_v	3	8
		24.

Proposition

$\mathrm{Aut}(C) \cong S_4$.

Proof

The cube can be viewed through its four body diagonals. Distinct rotations in $\mathrm{Aut}(C)$ induce different permutations of the four diagonals. Thus $\mathrm{Aut}(C) \cong S_4$.



The four body diagonals permuted by cube rotations.

Remark Relabeling the diagonals gives a different isomorphism with S_4. The resulting copies of S_4 are identified by conjugation; relabeling does not change cycle structure.

Remark For every group G, conjugation by any element,

$$g \longmapsto sgs^{-1},$$

is an automorphism of G.

Conjugacy Classes of $\mathrm{Aut}(C)$

# elements	label	S_4	$\mathrm{Aut}(C)$
1	$1A$	1	1
6	$2A$	(12)	rotations about edge-centered axes
3	$2B$	(12)(34)	rotations about face-centered axes by π
8	$3A$	(123)	rotations about vertex axes
6	$4A$	(1234)	rotations by $\pi/2$ or $3\pi/2$ about face centers.

For Cauchy–Frobenius, let V, E, F denote vertices, edges, and faces. It is enough to compute one representative from each conjugacy class:

class	# elements	FP_V	FP_E	FP_F	$\mathrm{FP}_{V\times V}$
$1A$	1	8	12	6	64
$2A$	6	0	2	0	0
$2B$	3	0	0	2	0
$3A$	8	2	0	0	4
$4A$	6	0	0	2	0.

Conjugate elements have the same number of fixed points, so one calculation for each conjugacy class suffices. The number of orbits of $\mathrm{Aut}(C)$ acting on $V \times V$ is

$$\frac{1}{|G|}\sum_{g\in G}\mathrm{FP}_{V\times V}(g) = \frac{1}{24}(64\cdot 1 + 8\cdot 4) = 4.$$

The four orbits in $V \times V$ are

$$\begin{aligned}
\mathcal{O}_1 &= \{(v, v)\}, \\
\mathcal{O}_2 &= \{(v_1, v_2) : (v_1, v_2) \text{ gives an edge}\}, \\
\mathcal{O}_3 &= \{(v_1, v_3) : (v_1, v_3) \text{ is a diagonal of a face}\}, \\
\mathcal{O}_4 &= \{(v_1, v_4) : (v_1, v_4) \text{ is a diagonal of the cube}\}.
\end{aligned}$$

Proposition

If G acts transitively on X and $|X| > 1$, then there is a $g \in G$ such that

$$\mathrm{FP}_X(g) = 0.$$

Proof

By Cauchy–Frobenius,

$$\begin{aligned}
1 &= \frac{1}{|G|} \sum_{g \in G} \mathrm{FP}_X(g) \\
&= \frac{1}{|G|} \left(\mathrm{FP}_X(1) + \sum_{g \in G \setminus \{1\}} \mathrm{FP}_X(g) \right) \\
&= \frac{1}{|G|} \left(|X| + \sum_{g \in G \setminus \{1\}} \mathrm{FP}_X(g) \right).
\end{aligned}$$

If $\mathrm{FP}_X(g) \geq 1$ for every $g \in G$, then the last quantity is greater than 1, a contradiction.

Theorem

Suppose G acts transitively on X, and

$$\mathrm{FP}_X(g) = 0 \qquad \text{for every } g \neq 1.$$

Then

$$X \cong G$$

as G-sets, viewing G as a G-set under multiplication $G \times G \to G$.

Definition

Such actions are called *free actions*. Examples include C_n acting on a regular n-gon and a group acting on itself by left multiplication.

Proof

By Cauchy–Frobenius and transitivity,

$$1 = \#\text{orbits} = \frac{1}{|G|} \sum_{g \in G} \mathrm{FP}_X(g) = \frac{|X|}{|G|},$$

so $|X| = |G|$. Choose $x \in X$ and define

$$\varphi : G \longrightarrow X, \qquad g \longmapsto gx.$$

Transitivity makes φ surjective. Since $|G| = |X|$, it is injective, and therefore bijective. It is compatible with the group action:

$$\varphi(gh) = (gh)x = g(hx) = g\varphi(h).$$

Example

$$\mathrm{Aff}(R) = \{T : R \to R : T(x) = ax + b,\ a \in R^*,\ b \in R\},$$

where R^* denotes the invertible elements of R.

For a ring R, $\mathrm{Aff}(R)$ is a group. Let

$$H = \{(1, b) : b \in R\} \cong (R, +), \qquad H \trianglelefteq \mathrm{Aff}(R),$$

and

$$H' = \{(a, 0) : a \in R^*\} \cong R^*.$$

Proposition

$$H \trianglelefteq \mathrm{Aff}(R).$$

Proof

Define

$$\pi : \mathrm{Aff}(R) \longrightarrow R^*, \qquad (a, b) \longmapsto a.$$

Then π is a group homomorphism and $\ker \pi = H$, so H is normal in $\mathrm{Aff}(R)$.

Conclusion. $\mathrm{Aff}(R) \approx R \times R^*$ contains $(R, +)$ as a normal subgroup, and

$$\mathrm{Aff}(R)/(R, +) \cong R^*.$$

Notation. $\mathrm{Aff}(\mathbb{Z}/p\mathbb{Z})$ is called the Frobenius group of order $p(p-1)$.

If $N \trianglelefteq G$, $H \leq G$, and $G/N \cong H$ naturally, then we say that G is the *semidirect product* of N and H, written

$$G = N \rtimes H = H \ltimes N,$$

with the bar on the nonnormal subgroup's side.

Remark

$$G = N \rtimes H \iff G = NH = \{nh : n \in N,\ h \in H\} \quad \text{and} \quad N \cap H = \{1\}.$$

Proof

Suppose $G = N \rtimes H$, so $N \trianglelefteq G$. The composite

$$H \overset{i}{\hookrightarrow} G \xrightarrow{\pi} G/N$$

(inclusion followed by projection) is an isomorphism. Thus $\pi|_H : H \xrightarrow{\sim} G/N$. Injectivity implies $H \cap N = \{1\}$, and surjectivity implies $NH = G$. The converse is similar.

Remark In the context of affine groups,

$$\mathrm{Aff}(R) = (R, +) \rtimes R^*.$$

Alternating Groups

Proposition

Every $g \in S_n$ can be written as a product of transpositions.

Proof

$$(ab) = (ab), \qquad (abc) = (ac)(ab), \qquad (abcd) = (ad)(ac)(ab), \qquad \dots$$

Proof (continued)

In general,

$$(a_1 a_2 \cdots a_k) = (a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_2).$$

The result for a product of permutations follows.

Theorem

Given $g \in S_n$, the parity (odd or even) of the number of transpositions in a decomposition of g is invariant.

Proof

Represent g as a matrix in $\mathrm{GL}_n(\mathbb{R})$ which permutes the standard basis $\{e_1, \dots, e_n\}$ of $\mathbb{R}^n$:

$$g(e_1, \dots, e_n) = (e_1, \dots, e_n) \begin{bmatrix} | & | & & | \\ ge_1 & ge_2 & \cdots & ge_n \\ | & | & & | \end{bmatrix}.$$

For example,

$$(12)(e_1, e_2, e_3) = (e_1, e_2, e_3) \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

where $e_1 \mapsto e_2$, $e_2 \mapsto e_1$, and $e_3 \mapsto e_3$. Hence there is a map

$$j : S_n \hookrightarrow \mathrm{GL}_n(\mathbb{R}).$$

Definition

The composite

$$\mathrm{sign} := \det \circ j : S_n \longrightarrow \{\pm 1\}$$

is a homomorphism called the *sign homomorphism*. If $\mathrm{sign}(g) = 1$, then g is an even permutation; if $\mathrm{sign}(g) = -1$, then g is an odd permutation.
This concludes the proof: no matter how a permutation is written as a product of transpositions, its parity—equivalently its image under the sign homomorphism (which cannot be both $+1$ and -1)—is fixed.

Remark A k-cycle is even if k is odd, and odd if k is even.

Definition

The collection of even permutations in S_n is the alternating group A_n.

Proposition

$$A_n \trianglelefteq S_n.$$

Proof

$$A_n = \ker(\mathrm{sign}).$$

Equivalently, it has index 2 and is therefore normal.